CHAPTER 8

General Linear Transformations

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Introduction

In earlier sections we studied linear transformations from R^n to R^m. In this chapter we will define and study linear transformations from a general vector space V to a general vector space W. The results we will obtain here have important applications in physics, engineering, and various branches of mathematics.

8.1 General Linear Transformations

Up to now our study of linear transformations has focused on transformations from R^n to R^m. In this section we will turn our attention to linear transformations involving general vector spaces. We will illustrate ways in which such transformations arise, and we will establish a fundamental relationship between general n-dimensional vector spaces and R^n.

Definitions and Terminology

In Section 1.8 we defined a *matrix transformation* $T_A : R^n \rightarrow R^m$ to be a mapping of the form

$$T_A(\mathbf{x}) = A\mathbf{x}$$

in which A is an $m \times n$ matrix. We subsequently established in Theorem 1.8.3 that the matrix transformations are precisely the *linear transformations* from R^n to R^m; that is, the transformations with the linearity properties

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \quad \text{and} \quad T(k\mathbf{u}) = kT(\mathbf{u})$$

We will use these two properties as the starting point for defining more general linear transformations.

Definition 1

If $T : V \to W$ is a mapping from a vector space V to a vector space W, then T is called a ***linear transformation*** from V to W if the following two properties hold for all vectors $\mathbf{u}$ and $\mathbf{v}$ in V and for all scalars k:

(i) $T(k\mathbf{u}) = kT(\mathbf{u})$ **[Homogeneity property]**

(ii) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ **[Additivity property]**

In the special case where $V = W$, the linear transformation T is called a ***linear operator*** on the vector space V.

The homogeneity and additivity properties of a linear transformation $T : V \to W$ can be used in combination to show that if $\mathbf{v}_1$ and $\mathbf{v}_2$ are vectors in V and k_1 and k_2 are any scalars, then

$$T(k_1\mathbf{v}_1 + k_2\mathbf{v}_2) = k_1T(\mathbf{v}_1) + k_2T(\mathbf{v}_2)$$

More generally, if $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are vectors in V and $k_1, k_2, \ldots, k_n$ are any scalars, then

$$\boxed{T(k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \cdots + k_n\mathbf{v}_n) = k_1T(\mathbf{v}_1) + k_2T(\mathbf{v}_2) + \cdots + k_nT(\mathbf{v}_n)} \tag{1}$$

The following theorem is an analog of parts (a) and (d) of Theorem 1.8.1.

Theorem 8.1.1

If $T : V \to W$ is a linear transformation, then:

(a) $T(\mathbf{0}) = \mathbf{0}$.

(b) $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$ for all $\mathbf{u}$ and $\mathbf{v}$ in V.

(c) $T(-\mathbf{v}) = -T(\mathbf{v})$ for all $\mathbf{v}$ in V.

Proof Let $\mathbf{u}$ be any vector in V. Since $0\mathbf{u} = \mathbf{0}$, it follows from the homogeneity property in Definition 1 that

$$T(\mathbf{0}) = T(0\mathbf{u}) = 0T(\mathbf{u}) = \mathbf{0}$$

which proves (a). We can prove part (b) by rewriting $T(\mathbf{u} - \mathbf{v})$ as

$$\begin{aligned} T(\mathbf{u} - \mathbf{v}) &= T\big(\mathbf{u} + (-1)\mathbf{v}\big) \\ &= T(\mathbf{u}) + (-1)T(\mathbf{v}) \\ &= T(\mathbf{u}) - T(\mathbf{v}) \end{aligned}$$

We leave it for you to justify each step. To prove part (c) set $\mathbf{u} = \mathbf{0}$ in part (b) and apply part (a). ■

EXAMPLE 1 | Matrix Transformations

Because we have based the definition of a general linear transformation on the homogeneity and additivity properties of *matrix transformations*, it follows that every matrix transformation $T_A : R^n \to R^m$ is a linear transformation in the sense of Definition 1.

EXAMPLE 2 | The Zero Transformation

Let V and W be any two vector spaces. The mapping $T : V \to W$ defined by $T(\mathbf{v}) = \mathbf{0}$ for every $\mathbf{v}$ in V is a linear transformation called the ***zero transformation***. To see that T is linear, observe that

$$T(\mathbf{u} + \mathbf{v}) = \mathbf{0}, \quad T(\mathbf{u}) = \mathbf{0}, \quad T(\mathbf{v}) = \mathbf{0}, \quad \text{and} \quad T(k\mathbf{u}) = \mathbf{0}$$

Therefore,

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \quad \text{and} \quad T(k\mathbf{u}) = kT(\mathbf{u})$$

EXAMPLE 3 | The Identity Operator

Let V be any vector space. The mapping $I : V \to V$ defined by $I(\mathbf{v}) = \mathbf{v}$ is called the ***identity operator*** on V. We will leave it for you to verify that I is linear.

EXAMPLE 4 | Dilation and Contraction Operators

If V is a vector space and c is any scalar, then the linear operator $T : V \to V$ that is defined by $T(\mathbf{x}) = c\mathbf{x}$ is a linear operator on V, for if c is any scalar and if $\mathbf{u}$ and $\mathbf{v}$ are any vectors in V, then

$$T(k\mathbf{u}) = c(k\mathbf{u}) = k(c\mathbf{u}) = kT(\mathbf{u})$$
$$T(\mathbf{u} + \mathbf{v}) = c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v} = T(\mathbf{u}) + T(\mathbf{v})$$

If $0 < c < 1$, then T is called the ***contraction*** of V with factor c, and if $c > 1$, it is called the ***dilation*** of V with factor c.

EXAMPLE 5 | A Linear Transformation from P_n to P_{n+1}

Let $\mathbf{p} = p(x) = c_0 + c_1x + \cdots + c_nx^n$ be a polynomial in P_n, and define the transformation $T : P_n \to P_{n+1}$ by

$$T(\mathbf{p}) = T(p(x)) = xp(x) = c_0x + c_1x^2 + \cdots + c_nx^{n+1}$$

This transformation is linear because for any scalar k and any polynomials $\mathbf{p}_1$ and $\mathbf{p}_2$ in P_n we have

$$T(k\mathbf{p}) = T(kp(x)) = x(kp(x)) = k(xp(x)) = kT(\mathbf{p})$$

and

$$\begin{aligned} T(\mathbf{p}_1 + \mathbf{p}_2) &= T(p_1(x) + p_2(x)) = x(p_1(x) + p_2(x)) \\ &= xp_1(x) + xp_2(x) = T(\mathbf{p}_1) + T(\mathbf{p}_2) \end{aligned}$$

EXAMPLE 6 | A Linear Transformation Using an Inner Product

Let $\mathbf{v}_0$ be any fixed vector in a real inner product space V, and let $T : V \to R$ be the transformation

$$T(\mathbf{x}) = \langle \mathbf{x}, \mathbf{v}_0 \rangle$$

that maps a vector $\mathbf{x}$ to its inner product with $\mathbf{v}_0$. This transformation is linear, for if k is any scalar, and if $\mathbf{u}$ and $\mathbf{v}$ are any vectors in R^n, then it follows from properties of real inner products that

$$T(k\mathbf{u}) = \langle k\mathbf{u}, \mathbf{v}_0 \rangle = k\langle \mathbf{u}, \mathbf{v}_0 \rangle = kT(\mathbf{u})$$
$$T(\mathbf{u} + \mathbf{v}) = \langle \mathbf{u} + \mathbf{v}, \mathbf{v}_0 \rangle = \langle \mathbf{u}, \mathbf{v}_0 \rangle + \langle \mathbf{v}, \mathbf{v}_0 \rangle = T(\mathbf{u}) + T(\mathbf{v})$$

EXAMPLE 7 | Transformations on Matrix Spaces

Let M_{nn} be the vector space of $n \times n$ matrices. In each part determine whether the transformation is linear.

(*a*) $T_1(A) = A^T$ (*b*) $T_2(A) = \det(A)$

Solution (*a*) It follows from parts (*b*) and (*d*) of Theorem 1.4.8 that

$$T_1(kA) = (kA)^T = kA^T = kT_1(A)$$
$$T_1(A + B) = (A + B)^T = A^T + B^T = T_1(A) + T_1(B)$$

so T_1 is linear.

Solution (*b*) It follows from Formula (1) of Section 2.3 that

$$T_2(kA) = \det(kA) = k^n \det(A) = k^n T_2(A)$$

Thus, T_2 is not homogeneous and hence not linear if $n > 1$. Note that additivity also fails because we showed in Example 1 of Section 2.3 that $\det(A + B)$ and $\det(A) + \det(B)$ are not generally equal.

EXAMPLE 8 | Translation Is Not Linear

Part (*a*) of Theorem 8.1.1 states that a linear transformation maps $\mathbf{0}$ to $\mathbf{0}$. This property is useful for identifying transformations that are *not* linear. For example, if $\mathbf{x}_0$ is a fixed nonzero vector in a real inner product space V, then the transformation

$$T(\mathbf{x}) = \mathbf{x} + \mathbf{x}_0$$

has the geometric effect of translating each point $\mathbf{x}$ in a direction parallel to $\mathbf{x}_0$ through a distance of $\|\mathbf{x}_0\|$ (Figure 8.1.1). This cannot be a linear transformation since $T(\mathbf{0}) = \mathbf{x}_0$, so T does not map $\mathbf{0}$ to $\mathbf{0}$.

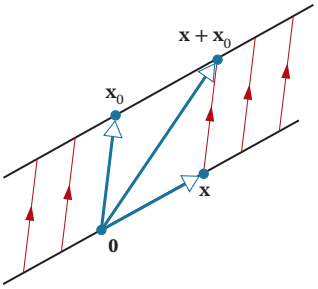


FIGURE 8.1.1 $T(\mathbf{x}) = \mathbf{x} + \mathbf{x}_0$ translates each point $\mathbf{x}$ along a line parallel to $\mathbf{x}_0$ through a distance $\|\mathbf{x}_0\|$.

EXAMPLE 9 | The Evaluation Transformation

Let V be a subspace of $F(-\infty, \infty)$, let

$$x_1, x_2, \ldots, x_n$$

be a sequence of distinct real numbers, and let $T : V \to R^n$ be the transformation

$$T(f) = \big(f(x_1), f(x_2), \ldots, f(x_n)\big) \tag{2}$$

that associates with the function f the n-tuple of function values at $x_1, x_2, \ldots, x_n$. We call this the ***evaluation transformation*** on V at $x_1, x_2, \ldots, x_n$. Thus, for example, if

$$x_1 = -1, \quad x_2 = 2, \quad x_3 = 4$$

and if $f(x) = x^2 - 1$, then

$$T(f) = \big(f(x_1), f(x_2), f(x_3)\big) = (0, 3, 15)$$

The evaluation transformation in (2) is linear, for if k is any scalar, and if f and g are any functions in V, then

$$\begin{aligned} T(kf) &= \big((kf)(x_1), (kf)(x_2), \ldots, (kf)(x_n)\big) \\ &= \big(kf(x_1), kf(x_2), \ldots, kf(x_n)\big) \\ &= k\big(f(x_1), f(x_2), \ldots, f(x_n)\big) = kT(f) \end{aligned}$$

and

$$\begin{aligned} T(f+g) &= \big((f+g)(x_1), (f+g)(x_2), \ldots, (f+g)(x_n)\big) \\ &= \big(f(x_1) + g(x_1), f(x_2) + g(x_2), \ldots, f(x_n) + g(x_n)\big) \\ &= \big(f(x_1), f(x_2), \ldots, f(x_n)\big) + \big(g(x_1), g(x_2), \ldots, g(x_n)\big) \\ &= T(f) + T(g) \end{aligned}$$

Finding Linear Transformations from Images of Basis Vectors

We saw in Formula (15) of Section 1.8 that if $T : R^n \to R^m$ is a linear transformation, and if $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n$ are the standard basis vectors for R^n, then the matrix A for T can be expressed as

$$A = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)]$$

It follows from this that the image of any vector $\mathbf{v} = (c_1, c_2, \ldots, c_n)$ in R^n under multiplication by A can be expressed as

$$T(\mathbf{v}) = c_1 T(\mathbf{e}_1) + c_2 T(\mathbf{e}_2) + \cdots + c_n T(\mathbf{e}_n)$$

This formula tells us that for a matrix transformation the image of any vector is expressible as a linear combination of the images of the standard basis vectors. This is a special case of the following more general result.

Theorem 8.1.2

Let $T : V \to W$ be a linear transformation, for which the vector space V is finite-dimensional. If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis for V, then the image of any vector $\mathbf{v}$ in V can be expressed as

$$T(\mathbf{v}) = c_1 T(\mathbf{v}_1) + c_2 T(\mathbf{v}_2) + \cdots + c_n T(\mathbf{v}_n) \tag{3}$$

where $c_1, c_2, \ldots, c_n$ are the coefficients required to express $\mathbf{v}$ as a linear combination of the vectors in the basis S.

Proof Express $\mathbf{v}$ as $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$ and use the linearity of T. ■

EXAMPLE 10 | Computing with Images of Basis Vectors

Consider the basis $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for R^3, where

$$\mathbf{v}_1 = (1,1,1), \quad \mathbf{v}_2 = (1,1,0), \quad \mathbf{v}_3 = (1,0,0)$$

Let $T : R^3 \to R^2$ be the linear transformation for which

$$T(\mathbf{v}_1) = (1,0), \quad T(\mathbf{v}_2) = (2,-1), \quad T(\mathbf{v}_3) = (4,3)$$

Find a formula for $T(x_1, x_2, x_3)$, and then use that formula to compute $T(2,-3,5)$.

Solution We first need to express $\mathbf{x} = (x_1, x_2, x_3)$ as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$. If we write

$$(x_1, x_2, x_3) = c_1(1,1,1) + c_2(1,1,0) + c_3(1,0,0)$$

then on equating corresponding components, we obtain

$$\begin{aligned} c_1 + c_2 + c_3 &= x_1 \\ c_1 + c_2 &= x_2 \\ c_1 &= x_3 \end{aligned}$$

which yields $c_1 = x_3$, $c_2 = x_2 - x_3$, $c_3 = x_1 - x_2$, so

$$\begin{aligned} (x_1, x_2, x_3) &= x_3(1,1,1) + (x_2 - x_3)(1,1,0) + (x_1 - x_2)(1,0,0) \\ &= x_3\mathbf{v}_1 + (x_2 - x_3)\mathbf{v}_2 + (x_1 - x_2)\mathbf{v}_3 \end{aligned}$$

Thus

$$\begin{aligned} T(x_1, x_2, x_3) &= x_3T(\mathbf{v}_1) + (x_2 - x_3)T(\mathbf{v}_2) + (x_1 - x_2)T(\mathbf{v}_3) \\ &= x_3(1,0) + (x_2 - x_3)(2,-1) + (x_1 - x_2)(4,3) \\ &= (4x_1 - 2x_2 - x_3, 3x_1 - 4x_2 + x_3) \end{aligned}$$

From this formula we obtain

$$T(2,-3,5) = (9,23)$$

EXAMPLE 11 | A Linear Transformation from $C^1(-\infty, \infty)$ to $F(-\infty, \infty)$

CALCULUS REQUIRED

Let $V = C^1(-\infty, \infty)$ be the vector space of functions with continuous first derivatives on $(-\infty, \infty)$, and let $W = F(-\infty, \infty)$ be the vector space of all real-valued functions defined on $(-\infty, \infty)$. Let $D : V \to W$ be the transformation that maps a function $\mathbf{f} = f(x)$ into its derivative—that is,

$$D(\mathbf{f}) = f'(x)$$

From the properties of differentiation, we have

$$D(\mathbf{f} + \mathbf{g}) = D(\mathbf{f}) + D(\mathbf{g}) \quad \text{and} \quad D(k\mathbf{f}) = kD(\mathbf{f})$$

Thus, D is a linear transformation.

CALCULUS REQUIRED

EXAMPLE 12 | An Integral Transformation

Let $V = C(-\infty, \infty)$ be the vector space of continuous functions on the interval $(-\infty, \infty)$, let $W = C^1(-\infty, \infty)$ be the vector space of functions with continuous first derivatives on $(-\infty, \infty)$, and let $J : V \to W$ be the transformation that maps a function f in V into

$$J(f) = \int_0^x f(t)\,dt$$

For example, if $f(x) = x^2$, then

$$J(f) = \int_0^x t^2\,dt = \frac{t^3}{3}\Big]_0^x = \frac{x^3}{3}$$

The transformation $J : V \to W$ is linear, for if k is any constant, and if f and g are any functions in V, then properties of the integral imply that

$$J(kf) = \int_0^x kf(t)\,dt = k\int_0^x f(t)\,dt = kJ(f)$$

$$J(f+g) = \int_0^x (f(t)+g(t))\,dt = \int_0^x f(t)\,dt + \int_0^x g(t)\,dt = J(f) + J(g)$$

Kernel and Range

Recall that if A is an $m \times n$ matrix, then the null space of A consists of all vectors $\mathbf{x}$ in R^n such that $A\mathbf{x} = \mathbf{0}$, and by Theorem 4.8.1 the column space of A consists of all vectors $\mathbf{b}$ in R^m for which there is at least one vector $\mathbf{x}$ in R^n such that $A\mathbf{x} = \mathbf{b}$. From the viewpoint of matrix transformations, the null space of A consists of all vectors in R^n that multiplication by A maps into $\mathbf{0}$, and the column space of A consists of all vectors in R^m that are images of at least one vector in R^n under multiplication by A. The following definition extends these ideas to general linear transformations, which is illustrated in (**Figure 8.1.2**).

Definition 2

If $T : V \to W$ is a linear transformation, then the set of vectors in V that T maps into $\mathbf{0}$ is called the ***kernel*** of T and is denoted by $\ker(T)$. The set of all vectors in W that are images under T of at least one vector in V is called the ***range*** of T and is denoted by $R(T)$.

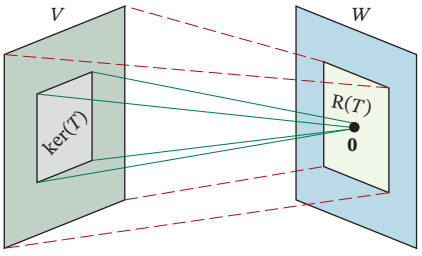


FIGURE 8.1.2

EXAMPLE 13 | Kernel and Range of a Matrix Transformation

If $T_A : R^n \rightarrow R^m$ is multiplication by the $m \times n$ matrix A, then the kernel of T_A is the null space of A, and the range of T_A is the column space of A.

EXAMPLE 14 | Kernel and Range of the Zero Transformation

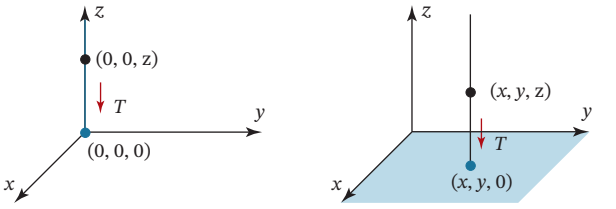
Let $T : V \rightarrow W$ be the zero transformation. Since T maps *every* vector in V into $\mathbf{0}$, it follows that $\ker(T) = V$. Moreover, since $\mathbf{0}$ is the *only* image under T of vectors in V, it follows that $R(T) = \{\mathbf{0}\}$.

EXAMPLE 15 | Kernel and Range of the Identity Operator

Let $I : V \rightarrow V$ be the identity operator. Since $I(\mathbf{v}) = \mathbf{v}$ for all vectors in V, *every* vector in V is the image of some vector (namely, itself); thus $R(I) = V$. Since the *only* vector that I maps into $\mathbf{0}$ is $\mathbf{0}$, it follows that $\ker(I) = \{\mathbf{0}\}$.

EXAMPLE 16 | Kernel and Range of an Orthogonal Projection

Let $T : R^3 \rightarrow R^3$ be the orthogonal projection onto the xy-plane. As illustrated in **Figure 8.1.3*a***, the points that T maps into $\mathbf{0} = (0, 0, 0)$ are precisely those on the z-axis, so $\ker(T)$ is the set of points of the form $(0, 0, z)$. As illustrated in **Figure 8.1.3*b***, T maps the points in R^3 to the xy-plane, where each point in that plane is the image of each point on the vertical line above it. Thus, $R(T)$ is the set of points of the form $(x, y, 0)$.



(*a*) $\ker(T)$ is the z-axis. (*b*) $R(T)$ is the entire xy-plane.

FIGURE 8.1.3

EXAMPLE 17 | Kernel and Range of a Rotation

Let $T : R^2 \rightarrow R^2$ be the linear operator that rotates each vector in the xy-plane through the angle θ (**Figure 8.1.4**). Since *every* vector in the xy-plane can be obtained by rotating some vector through the angle θ, it follows that $R(T) = R^2$. Moreover, the only vector that rotates into $\mathbf{0}$ is $\mathbf{0}$, so $\ker(T) = \{\mathbf{0}\}$.

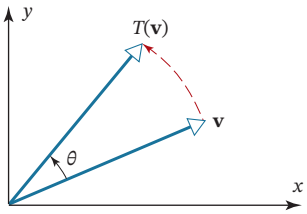


FIGURE 8.1.4

CALCULUS REQUIRED

EXAMPLE 18 | Kernel of a Differentiation Transformation

Let $V = C^1(-\infty, \infty)$ be the vector space of functions with continuous first derivatives on $(-\infty, \infty)$, let $W = F(-\infty, \infty)$ be the vector space of all real-valued functions defined on $(-\infty, \infty)$, and let $D : V \to W$ be the differentiation transformation $D(\mathbf{f}) = f'(x)$. The kernel of D is the set of functions in V with derivative zero. As shown in calculus, this is the set of constant functions on $(-\infty, \infty)$.

Properties of Kernel and Range

In all of the preceding examples, $\ker(T)$ and $R(T)$ turned out to be *subspaces*. In Examples 14, 15, and 17 they were either the zero subspace or the entire vector space. In Example 16 the kernel was a line through the origin, and the range was a plane through the origin, both of which are subspaces of R^3. All of this is a consequence of the following general theorem.

Theorem 8.1.3

If $T : V \to W$ is a linear transformation, then:

(a) The kernel of T is a subspace of V.

(b) The range of T is a subspace of W.

Proof (a) To show that $\ker(T)$ is a subspace, we must show that it contains at least one vector and is closed under addition and scalar multiplication. By part (a) of Theorem 8.1.1, the vector $\mathbf{0}$ is in $\ker(T)$, so the kernel contains at least one vector. If $\mathbf{0}$ is the only vector in the kernel of T, then $\ker(T)$ is the zero subspace of V. If there are at least two vectors in the kernel, then let $\mathbf{v}_1$ and $\mathbf{v}_2$ be any two such vectors, and let k be any scalar. Then

$$T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2) = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

so $\mathbf{v}_1 + \mathbf{v}_2$ is in $\ker(T)$. Also,

$$T(k\mathbf{v}_1) = kT(\mathbf{v}_1) = k\mathbf{0} = \mathbf{0}$$

so $k\mathbf{v}_1$ is in $\ker(T)$.

Proof (b) To show that $R(T)$ is a subspace of W, we must show that it contains at least one vector and is closed under addition and scalar multiplication. However, it contains at least the zero vector of W since $T(\mathbf{0}) = \mathbf{0}$ by part (a) of Theorem 8.1.1. To prove that it is closed under addition and scalar multiplication, we must show that if $\mathbf{w}_1$ and $\mathbf{w}_2$ are vectors in $R(T)$, and if k is any scalar, then there exist vectors $\mathbf{a}$ and $\mathbf{b}$ in V for which

$$T(\mathbf{a}) = \mathbf{w}_1 + \mathbf{w}_2 \quad \text{and} \quad T(\mathbf{b}) = k\mathbf{w}_1 \tag{4}$$

But the fact that $\mathbf{w}_1$ and $\mathbf{w}_2$ are in $R(T)$ tells us there exist vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ in V such that

$$T(\mathbf{v}_1) = \mathbf{w}_1 \quad \text{and} \quad T(\mathbf{v}_2) = \mathbf{w}_2$$

The following computations complete the proof by showing that the vectors $\mathbf{a} = \mathbf{v}_1 + \mathbf{v}_2$ and $\mathbf{b} = k\mathbf{v}_1$ satisfy the equations in (4):

$$T(\mathbf{a}) = T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2) = \mathbf{w}_1 + \mathbf{w}_2$$
$$T(\mathbf{b}) = T(k\mathbf{v}_1) = kT(\mathbf{v}_1) = k\mathbf{w}_1 \quad \blacksquare$$

EXAMPLE 19 | Application to Differential Equations

CALCULUS REQUIRED

Differential equations of the form

$$y'' + \omega^2 y = 0 \qquad (\omega \text{ a positive constant}) \tag{5}$$

arise in the study of vibrations. The set of all solutions of this equation on the interval $(-\infty, \infty)$ is the kernel of the linear transformation $D : C^2(-\infty, \infty) \to C(-\infty, \infty)$, given by

$$D(y) = y'' + \omega^2 y$$

It is proved in standard textbooks on differential equations that the kernel is a two-dimensional subspace of $C^2(-\infty, \infty)$, so that if we can find two linearly independent solutions of (5), then all other solutions can be expressed as linear combinations of those two. We leave it for you to confirm by differentiating that

$$y_1 = \cos \omega x \quad \text{and} \quad y_2 = \sin \omega x$$

are solutions of (5). These functions are linearly independent since neither is a scalar multiple of the other, and thus

$$y = c_1 \cos \omega x + c_2 \sin \omega x \tag{6}$$

is a "general solution" of (5) in the sense that every choice of c_1 and c_2 produces a solution, and every solution is of this form.

Rank and Nullity of Linear Transformations

In Definition 1 of Section 4.9 we defined the notions of *rank* and *nullity* for an $m \times n$ matrix, and in Theorem 4.9.2, which we called the *Dimension Theorem for Matrices*, we proved that the sum of the rank and nullity is n. We will show next that this result is a special case of a more general result about linear transformations. We start with the following definition.

Definition 3

Let $T : V \to W$ be a linear transformation. In the case that the range of T is finite-dimensional its dimension is called the ***rank of T***; and if the kernel of T is finite-dimensional, then its dimension is called the ***nullity of T***. These dimensions are denoted, respectively, by

$$\text{rank}(T) \quad \text{and} \quad \text{nullity}(T)$$

The following theorem, whose proof is optional, generalizes Theorem 4.9.2.

Theorem 8.1.4

Dimension Theorem for Linear Transformations

If $T : V \to W$ is a linear transformation from a finite-dimensional vector space V to a vector space W, then the range of T is finite-dimensional, and

$$\text{rank}(T) + \text{nullity}(T) = \dim(V) \tag{7}$$

In the special case where A is an $m \times n$ matrix and $T_A : R^n \to R^m$ is multiplication by A, the kernel of T_A is the null space of A, and the range of T_A is the column space of A. Thus, it follows from Theorem 8.1.4 that

$$\text{rank}(T_A) + \text{nullity}(T_A) = n$$

OPTIONAL: Proof of Theorem 8.1.4 Assume that V is n-dimensional. We must show that

$$\dim(R(T)) + \dim(\ker(T)) = n$$

We will give the proof for the case where $1 \leq \dim(\ker(T)) < n$. The cases where $\dim(\ker(T)) = 0$ and $\dim(\ker(T)) = n$ are left as exercises. Assume $\dim(\ker(T)) = r$, and let $\mathbf{v}_1, \ldots, \mathbf{v}_r$ be a basis for the kernel. Since $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is linearly independent, Theorem 4.6.5(b) states that there are $n - r$ vectors, $\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n$, such that the extended set $\{\mathbf{v}_1, \ldots, \mathbf{v}_r, \mathbf{v}_{r+1}, \ldots, \mathbf{v}_n\}$ is a basis for V. To complete the proof, we will show that the $n - r$ vectors in the set $S = \{T(\mathbf{v}_{r+1}), \ldots, T(\mathbf{v}_n)\}$ form a basis for the range of T. It will then follow that

$$\dim(R(T)) + \dim(\ker(T)) = (n - r) + r = n$$

First we show that S spans the range of T. If $\mathbf{b}$ is any vector in the range of T, then $\mathbf{b} = T(\mathbf{v})$ for some vector $\mathbf{v}$ in V. Since $\{\mathbf{v}_1, \ldots, \mathbf{v}_r, \mathbf{v}_{r+1}, \ldots, \mathbf{v}_n\}$ is a basis for V, the vector $\mathbf{v}$ can be written in the form

$$\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_r\mathbf{v}_r + c_{r+1}\mathbf{v}_{r+1} + \cdots + c_n\mathbf{v}_n$$

Since $\mathbf{v}_1, \ldots, \mathbf{v}_r$ lie in the kernel of T, we have $T(\mathbf{v}_1) = \cdots = T(\mathbf{v}_r) = \mathbf{0}$, so

$$\mathbf{b} = T(\mathbf{v}) = c_{r+1}T(\mathbf{v}_{r+1}) + \cdots + c_nT(\mathbf{v}_n)$$

Thus S spans the range of T.

Finally, we show that S is a linearly independent set and consequently forms a basis for the range of T. Suppose that some linear combination of the vectors in S is zero; that is,

$$k_{r+1}T(\mathbf{v}_{r+1}) + \cdots + k_nT(\mathbf{v}_n) = \mathbf{0} \tag{8}$$

We must show that $k_{r+1} = \cdots = k_n = 0$. Since T is linear, (8) can be rewritten as

$$T(k_{r+1}\mathbf{v}_{r+1} + \cdots + k_n\mathbf{v}_n) = \mathbf{0}$$

which says that $k_{r+1}\mathbf{v}_{r+1} + \cdots + k_n\mathbf{v}_n$ is in the kernel of T. This vector can therefore be written as a linear combination of the basis vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$, say

$$k_{r+1}\mathbf{v}_{r+1} + \cdots + k_n\mathbf{v}_n = k_1\mathbf{v}_1 + \cdots + k_r\mathbf{v}_r$$

Thus,

$$k_1\mathbf{v}_1 + \cdots + k_r\mathbf{v}_r - k_{r+1}\mathbf{v}_{r+1} - \cdots - k_n\mathbf{v}_n = \mathbf{0}$$

Since $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly independent, all of the k's are zero; in particular, $k_{r+1} = \cdots = k_n = 0$, which completes the proof. ■

Exercise Set 8.1

In Exercises 1–2, suppose that T is a mapping whose domain is the vector space M_{22}. In each part, determine whether T is a linear transformation, and if so, find its kernel.

1. **a.** $T(A) = A^2$ **b.** $T(A) = \text{tr}(A)$
   c. $T(A) = A + A^T$

2. **a.** $T(A) = (A)_{11}$ **b.** $T(A) = 0_{2\times 2}$
   c. $T(A) = cA$

In Exercises 3–9, determine whether the mapping T is a linear transformation, and if so, find its kernel.

3. $T : R^3 \to R$, where $T(\mathbf{u}) = \|\mathbf{u}\|$.

4. $T : R^3 \to R^3$, where $\mathbf{v}_0$ is a fixed vector in R^3 and
$$T(\mathbf{u}) = \mathbf{u} \times \mathbf{v}_0$$

5. $T : M_{22} \to M_{23}$, where B is a fixed 2×3 matrix and
$$T(A) = AB$$

6. $T : M_{22} \to R$, where
   a. $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = 3a - 4b + c - d$
   b. $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a^2 + b^2$

7. $T : P_2 \to P_2$, where
   a. $T(a_0 + a_1x + a_2x^2) = a_0 + a_1(x + 1) + a_2(x + 1)^2$
   b. $T(a_0 + a_1x + a_2x^2) = (a_0 + 1) + (a_1 + 1)x + (a_2 + 1)x^2$

8. $T : F(-\infty, \infty) \to F(-\infty, \infty)$, where
   a. $T(f(x)) = 1 + f(x)$ **b.** $T(f(x)) = f(x + 1)$

9. $T : R^\infty \to R^\infty$, where

$$T(a_0, a_1, a_2, \ldots, a_n, \ldots) = (0, a_0, a_1, a_2, \ldots, a_n, \ldots)$$

10. Let $T : P_2 \to P_3$ be the linear transformation defined by $T(p(x)) = xp(x)$. Which of the following are in $\ker(T)$?

a. x^2 **b.** 0 **c.** $1 + x$ **d.** $-x$

11. Let $T : P_2 \to P_3$ be the linear transformation in Exercise 10. Which of the following are in $R(T)$?

a. $x + x^2$ **b.** $1 + x$ **c.** $3 - x^2$ **d.** $-x$

12. Let V be any vector space, and let $T : V \to V$ be defined by $T(\mathbf{v}) = 3\mathbf{v}$.

a. What is the kernel of T?

b. What is the range of T?

13. In each part, use the given information to find the nullity of the linear transformation T.

a. $T : R^5 \to P_5$ has rank 3.

b. $T : P_4 \to P_3$ has rank 1.

c. The range of $T : M_{mn} \to R^3$ is R^3.

d. $T : M_{22} \to M_{22}$ has rank 3.

14. In each part, use the given information to find the rank of the linear transformation T.

a. $T : R^7 \to M_{32}$ has nullity 2.

b. $T : P_3 \to R$ has nullity 1.

c. The null space of $T : P_5 \to P_5$ is P_5.

d. $T : P_n \to M_{mn}$ has nullity 3.

15. Let $T : M_{22} \to M_{22}$ be the dilation operator with factor $k = 3$.

a. Find $T\left(\begin{bmatrix} 1 & 2 \\ -4 & 3 \end{bmatrix}\right)$.

b. Find the rank and nullity of T.

16. Let $T : P_2 \to P_2$ be the contraction operator with factor $k = 1/4$.

a. Find $T(1 + 4x + 8x^2)$.

b. Find the rank and nullity of T.

17. Let $T : P_2 \to R^3$ be the evaluation transformation at the sequence of points $-1, 0, 1$. Find

a. $T(x^2)$ **b.** $\ker(T)$ **c.** $R(T)$

18. Let V be the subspace of $C[0, 2\pi]$ spanned by the vectors 1, $\sin x$, and $\cos x$, and let $T : V \to R^3$ be the evaluation transformation at the sequence of points $0, \pi, 2\pi$. Find

a. $T(1 + \sin x + \cos x)$ **b.** $\ker(T)$

c. $R(T)$

19. Consider the basis $S = \{\mathbf{v}_1, \mathbf{v}_2\}$ for R^2, where $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (1, 0)$, and let $T : R^2 \to R^2$ be the linear operator for which

$$T(\mathbf{v}_1) = (1, -2) \quad \text{and} \quad T(\mathbf{v}_2) = (-4, 1)$$

Find a formula for $T(x_1, x_2)$, and use that formula to find $T(5, -3)$.

20. Consider the basis $S = \{\mathbf{v}_1, \mathbf{v}_2\}$ for R^2, where $\mathbf{v}_1 = (-2, 1)$ and $\mathbf{v}_2 = (1, 3)$, and let $T : R^2 \to R^3$ be the linear transformation such that

$$T(\mathbf{v}_1) = (-1, 2, 0) \quad \text{and} \quad T(\mathbf{v}_2) = (0, -3, 5)$$

Find a formula for $T(x_1, x_2)$, and use that formula to find $T(2, -3)$.

21. Consider the basis $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for R^3, where $\mathbf{v}_1 = (1, 1, 1)$, $\mathbf{v}_2 = (1, 1, 0)$, and $\mathbf{v}_3 = (1, 0, 0)$, and let $T : R^3 \to R^3$ be the linear operator for which

$$T(\mathbf{v}_1) = (2, -1, 4), \quad T(\mathbf{v}_2) = (3, 0, 1), \quad T(\mathbf{v}_3) = (-1, 5, 1)$$

Find a formula for $T(x_1, x_2, x_3)$, and use that formula to find $T(2, 4, -1)$.

22. Consider the basis $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for R^3, where $\mathbf{v}_1 = (1, 2, 1)$, $\mathbf{v}_2 = (2, 9, 0)$, and $\mathbf{v}_3 = (3, 3, 4)$, and let $T : R^3 \to R^2$ be the linear transformation for which

$$T(\mathbf{v}_1) = (1, 0), \quad T(\mathbf{v}_2) = (-1, 1), \quad T(\mathbf{v}_3) = (0, 1)$$

Find a formula for $T(x_1, x_2, x_3)$, and use that formula to find $T(7, 13, 7)$.

In Exercises 23–24, let T be multiplication by the matrix A. Find

a. a basis for the range of T.

b. a basis for the kernel of T.

c. the rank and nullity of T.

d. the rank and nullity of A.

23. $A = \begin{bmatrix} 1 & -1 & 3 \\ 5 & 6 & -4 \\ 7 & 4 & 2 \end{bmatrix}$ **24.** $A = \begin{bmatrix} 2 & 0 & -1 \\ 4 & 0 & -2 \\ 20 & 0 & 0 \end{bmatrix}$

In Exercises 25–26, let $T_A : R^4 \to R^3$ be multiplication by A. Find a basis for the kernel of T_A, and then find a basis for the range of T_A that consists of column vectors of A.

25. $A = \begin{bmatrix} 1 & 2 & -1 & -2 \\ -3 & 1 & 3 & 4 \\ -3 & 8 & 4 & 2 \end{bmatrix}$

26. $A = \begin{bmatrix} 1 & 1 & 0 & 1 \\ -2 & 4 & 2 & 2 \\ -1 & 8 & 3 & 5 \end{bmatrix}$

27. Let $T : P_3 \to P_2$ be the mapping defined by

$$T(a_0 + a_1x + a_2x^2 + a_3x^3) = 5a_0 + a_3x^2$$

a. Show that T is linear.

b. Find a basis for the kernel of T.

c. Find a basis for the range of T.

28. Let $T : P_2 \to P_2$ be the mapping defined by

$$T(a_0 + a_1x + a_2x^2) = 3a_0 + a_1x + (a_0 + a_1)x^2$$

a. Show that T is linear.

b. Find a basis for the kernel of T.

c. Find a basis for the range of T.

29. a. **(*Calculus required*)** Let $D : P_3 \to P_2$ be the differentiation transformation $D(\mathbf{p}) = p'(x)$. What is the kernel of D?

b. **(*Calculus required*)** Let $J : P_1 \to R$ be the integration transformation $J(\mathbf{p}) = \int_{-1}^{1} p(x)\,dx$. What is the kernel of J?

30. **(*Calculus required*)** Let $V = C[a, b]$ be the vector space of continuous functions on $[a, b]$, and let $T : V \to V$ be the transformation defined by

$$T(\mathbf{f}) = 5f(x) + 3\int_a^x f(t)\,dt$$

Is T a linear operator?

31. **(*Calculus required*)** Let V be the vector space of real-valued functions with continuous derivatives of all orders on the interval $(-\infty, \infty)$, and let $W = F(-\infty, \infty)$ be the vector space of real-valued functions defined on $(-\infty, \infty)$.

a. Find a linear transformation $T : V \to W$ whose kernel is P_3.

b. Find a linear transformation $T : V \to W$ whose kernel is P_n.

32. For a positive integer $n > 1$, let $T : M_{nn} \to R$ be the linear transformation defined by $T(A) = \text{tr}(A)$, where A is an $n \times n$ matrix with real entries. Determine the dimension of $\ker(T)$.

33. a. Let $T : V \to R^3$ be a linear transformation from a vector space V to R^3. Geometrically, what are the possibilities for the range of T?

b. Let $T : R^3 \to W$ be a linear transformation from R^3 to a vector space W. Geometrically, what are the possibilities for the kernel of T?

34. In each part, determine whether the mapping $T : P_n \to P_n$ is linear.

a. $T(p(x)) = p(x + 1)$

b. $T(p(x)) = p(x) + 1$

35. Let $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ be vectors in a vector space V, and let $T : V \to R^3$ be a linear transformation for which

$$T(\mathbf{v}_1) = (1, -1, 2), \quad T(\mathbf{v}_2) = (0, 3, 2),$$
$$T(\mathbf{v}_3) = (-3, 1, 2)$$

Find $T(2\mathbf{v}_1 - 3\mathbf{v}_2 + 4\mathbf{v}_3)$.

Working with Proofs

36. Let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a basis for a vector space V, and let $T : V \to W$ be a linear transformation. Prove that if

$$T(\mathbf{v}_1) = T(\mathbf{v}_2) = \cdots = T(\mathbf{v}_n) = \mathbf{0}$$

then T is the zero transformation.

37. Let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a basis for a vector space V, and let $T : V \to V$ be a linear operator. Prove that if

$$T(\mathbf{v}_1) = \mathbf{v}_1, \quad T(\mathbf{v}_2) = \mathbf{v}_2, \ldots, \quad T(\mathbf{v}_n) = \mathbf{v}_n$$

then T is the identity transformation on V.

38. Prove: If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis for a vector space V and $\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_n$ are vectors in a vector space W, not necessarily distinct, then there exists a linear transformation T that maps V into W such that

$$T(\mathbf{v}_1) = \mathbf{w}_1, \quad T(\mathbf{v}_2) = \mathbf{w}_2, \ldots, \quad T(\mathbf{v}_n) = \mathbf{w}_n$$

39. Let $q_0(x)$ be a fixed polynomial of degree m, and define a function T with domain P_n by the formula $T(p(x)) = p(q_0(x))$. Prove that T is a linear transformation.

True-False Exercises

TF. In parts **(a)–(i)** determine whether the statement is true or false, and justify your answer.

a. If $T(c_1\mathbf{v}_1 + c_2\mathbf{v}_2) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2)$ for all vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ in V and all scalars c_1 and c_2, then T is a linear transformation.

b. If $\mathbf{v}$ is a nonzero vector in V, then there is exactly one linear transformation $T : V \to W$ such that

$$T(-\mathbf{v}) = -T(\mathbf{v})$$

c. There is exactly one linear transformation $T : V \to W$ for which $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u} - \mathbf{v})$ for all vectors $\mathbf{u}$ and $\mathbf{v}$ in V.

d. If $\mathbf{v}_0$ is a nonzero vector in V, then $T(\mathbf{v}) = \mathbf{v}_0 + \mathbf{v}$ defines a linear operator on V.

e. The kernel of a linear transformation is a vector space.

f. The range of a linear transformation is a vector space.

g. If $T : P_6 \to M_{22}$ is a linear transformation, then the nullity of T is 3.

h. The function $T : M_{22} \to R$ defined by $T(A) = \det A$ is a linear transformation.

i. The linear transformation $T : M_{22} \to M_{22}$ defined by

$$T(A) = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} A$$

has rank 1.

8.2 Compositions and Inverse Transformations

In Section 1.9 we discussed compositions and inverses of matrix transformations. In this section we will extend some of those ideas to general linear transformations.

One-to-One and Onto

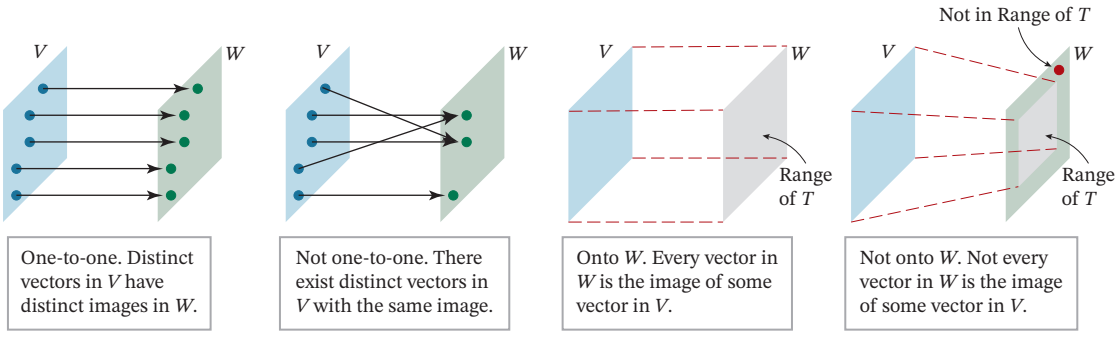
To set the groundwork for our discussion in this section we will need the following definitions that are illustrated in **Figure 8.2.1**.

Definition 1

If $T : V \to W$ is a linear transformation from a vector space V to a vector space W, then T is said to be ***one-to-one*** if T maps distinct vectors in V into distinct vectors in W.

Definition 2

If $T : V \to W$ is a linear transformation from a vector space V to a vector space W, then T is said to be ***onto*** (or ***onto W***) if every vector in W is the image of at least one vector in V.



One-to-one. Distinct vectors in V have distinct images in W.

Not one-to-one. There exist distinct vectors in V with the same image.

Onto W. Every vector in W is the image of some vector in V.

Not onto W. Not every vector in W is the image of some vector in V.

FIGURE 8.2.1

The idea of a one-to-one linear transformation can be expressed in other ways as well:

1. $T : V \to W$ is one-to-one if and only if for each vector $\mathbf{w}$ in the range of T, there is exactly one vector $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{w}$.
2. $T : V \to W$ is one-to-one if and only if $T(\mathbf{u}) = T(\mathbf{v})$ implies that $\mathbf{u} = \mathbf{v}$.

Recall from Definition 2 of Section 8.1 that the *kernel* of a linear transformation consists of all vectors that the transformation maps into $\mathbf{0}$. The following theorem links that definition with the concept of a one-to-one linear transformation.

Theorem 8.2.1

If $T : V \to W$ is a linear transformation, then the following two statements are equivalent.

(a) T is one-to-one.

(b) $\ker(T) = \{\mathbf{0}\}$.

Proof **$(a) \Rightarrow (b)$** Since T is linear, we know that $T(\mathbf{0}) = \mathbf{0}$ by Theorem 8.1.1(a). Since T is one-to-one, there can be no other vectors in V that map into $\mathbf{0}$, so $\ker(T) = \{\mathbf{0}\}$.

$(b) \Rightarrow (a)$ Assume that $\ker(T) = \{\mathbf{0}\}$. If $\mathbf{u}$ and $\mathbf{v}$ are distinct vectors in V, then $\mathbf{u} - \mathbf{v} \neq \mathbf{0}$. This implies that $T(\mathbf{u} - \mathbf{v}) \neq \mathbf{0}$, for otherwise $\ker(T)$ would contain a nonzero vector. Since T is linear, it follows that

$$T(\mathbf{u}) - T(\mathbf{v}) = T(\mathbf{u} - \mathbf{v}) \neq \mathbf{0}$$

so T maps distinct vectors in V into distinct vectors in W and hence is one-to-one. ■

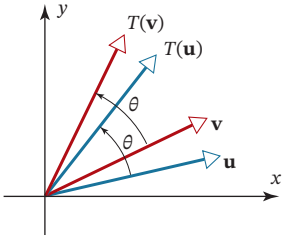


FIGURE 8.2.2 Distinct vectors $\mathbf{u}$ and $\mathbf{v}$ are rotated into distinct vectors $T(\mathbf{u})$ and $T(\mathbf{v})$.

EXAMPLE 1 | Rotation Operators on R^2 Are One-to-One and Onto

The linear operator $T : R^2 \to R^2$ that rotates each vector in the plane about the origin through an angle θ is one-to-one because it maps distinct vectors into distinct vectors (Figure 8.2.2). It is also onto because every vector in R^2 is the image under this rotation of another vector in R^2 (which vector?).

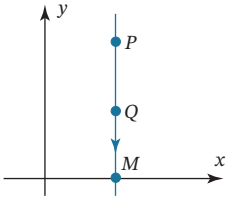


FIGURE 8.2.3 The distinct points P and Q are mapped into the same point M.

EXAMPLE 2 | Orthogonal Projections in R^2 Are Not One-to-One

The linear operator $T : R^2 \to R^2$ that maps points orthogonally on to the x-axis in R^2 maps distinct points on a vertical line to the same point on the x-axis and hence is not one-to-one (Figure 8.2.3). It is also not onto R^2 because points off the x-axis are not images of any point in R^2 under such a projection. Similarly, orthogonal projections onto the y-axis are neither one-to-one nor onto.

EXAMPLE 3 | Two Transformations That Are One-to-One and Onto

The linear transformations $T_1 : P_3 \to R^4$ and $T_2 : M_{22} \to R^4$ defined by

$$T_1(a + bx + cx^2 + dx^3) = (a, b, c, d)$$

$$T_2\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a, b, c, d)$$

are both onto R^4 because every vector in R^4 can be obtained by choosing a, b, c, and d appropriately. Both transformations are one-to-one because their kernels contain only the zero vector in their respective domains (verify).

EXAMPLE 4 | A One-to-One Linear Transformation That Is Not Onto

Let $T : P_n \to P_{n+1}$ be the linear transformation

$$T(\mathbf{p}) = T(p(x)) = xp(x)$$

discussed in Example 5 of Section 8.1. If

$$\mathbf{p} = p(x) = c_0 + c_1x + \cdots + c_nx^n \quad \text{and} \quad \mathbf{q} = q(x) = d_0 + d_1x + \cdots + d_nx^n$$

are distinct polynomials, then they differ in at least one coefficient, and hence

$$T(\mathbf{p}) = c_0x + c_1x^2 + \cdots + c_nx^{n+1} \quad \text{and} \quad T(\mathbf{q}) = d_0x + d_1x^2 + \cdots + d_nx^{n+1}$$

also differ in at least one coefficient. Thus, T is one-to-one, since it maps distinct polynomials into distinct polynomials. However, it is not onto because all images under T have a zero constant term, and hence there is no polynomial in P_n that maps into the constant polynomial 1.

EXAMPLE 5 | Shifting Operators

Let $V = R^\infty$ be the sequence space discussed in Example 3 of Section 4.1, and consider the linear "shifting operators" on V defined by

$$T_1(u_1, u_2, \ldots, u_n, \ldots) = (0, u_1, u_2, \ldots, u_n, \ldots)$$
$$T_2(u_1, u_2, \ldots, u_n, \ldots) = (u_2, u_3, \ldots, u_n, \ldots)$$

(*a*) Show that T_1 is one-to-one but not onto.

(*b*) Show that T_2 is onto but not one-to-one.

Solution (*a*) The operator T_1 is one-to-one because distinct sequences in R^∞ obviously have distinct images. This operator is not onto because no vector in R^∞ maps into the sequence $(1, 0, 0, \ldots, 0, \ldots)$, for example.

Solution (*b*) The operator T_2 is not one-to-one because, for example, the distinct vectors $(1, 0, 0, \ldots, 0, \ldots)$ and $(2, 0, 0, \ldots, 0, \ldots)$ both map into $(0, 0, 0, \ldots, 0, \ldots)$. This operator is onto because every possible sequence of real numbers can be obtained with an appropriate choice of the numbers $u_2, u_3, \ldots, u_n, \ldots$.

EXAMPLE 6 | Differentiation Is Not One-to-One

CALCULUS REQUIRED

Let

$$D : C^1(-\infty, \infty) \to F(-\infty, \infty)$$

be the differentiation transformation discussed in Example 11 of Section 8.1. This linear transformation is *not* one-to-one because it maps functions that differ by a constant into the same function. For example,

$$D(x^2) = D(x^2 + 1) = 2x$$

In the special case where V and W are finite-dimensional and have the same dimension, we can add a third statement to those in Theorem 8.2.1.

Theorem 8.2.2

If V and W are finite-dimensional vector spaces with the same dimension, and if $T : V \to W$ is a linear transformation, then the following statements are equivalent.

(a) T is one-to-one.

(b) $\ker(T) = \{\mathbf{0}\}$.

(c) T is onto [i.e., $R(T) = W$].

Why does Example 5 not violate Theorem 8.2.2?

Proof We already know that (a) and (b) are equivalent by Theorem 8.2.1, so it suffices to show that (b) and (c) are equivalent. We leave it for you to do this by assuming that $\dim(V) = n$ and applying Theorem 8.1.4. ■

The requirement in Theorem 8.2.2 that V and W have the same dimension is essential for the validity of the theorem. In the exercises we will ask you to prove the following facts for the case where they do not have the same dimension.

- If $\dim(W) < \dim(V)$, then T cannot be one-to-one.
- If $\dim(V) < \dim(W)$, then T cannot be onto.

Stated informally, if a linear transformation maps a "bigger" space to a "smaller" space, then some points in the "bigger" space must have the same image; and if a linear transformation maps a "smaller" space to a "bigger" space, then there must be points in the "bigger" space that are not images of any points in the "smaller" space.

In retrospect, had Theorem 8.2.2 been available prior to Example 3, it would have sufficed to show that the transformations were *either* one-to-one *or* onto since P_3 and M_{22} have the same dimension as R^4 (dimension 4).

Matrix Transformations Revisited

Let us return for the moment to matrix transformations and consider an example that illustrates the two results about dimension that followed Theorem 8.2.2.

EXAMPLE 7 | Matrix Transformations from R^n to R^m

If $T_A : R^n \to R^m$ is multiplication by an $m \times n$ matrix A, then it follows from the discussion immediately following the proof of Theorem 8.2.2 that T_A is not one-to-one if $m < n$ and not onto if $n < m$. In the case where $m = n$, whether or not T_A is one-to-one or onto depends on the rank of the matrix A. However, in the exercises we will ask you to show that if A is invertible, then T_A will be both one-to-one and onto.

The following theorem illustrates that it is the column vectors of a matrix A that determine whether the matrix transformation $T_A : R^n \to R^m$ is one-to-one or onto.

Theorem 8.2.3

If $T_A : R^n \rightarrow R^m$ is a matrix transformation, then

(*a*) T_A is one-to-one if and only if the columns of A are linearly independent.

(*b*) T_A is onto if and only if the columns of A span R^m.

Proof (a) It follows from Theorem 8.2.1 that T_A is one-to-one if and only if A has nullity 0, which is equivalent to saying that A has rank m (Theorem 4.9.2), which is equivalent to saying that the m column vectors of A are linearly independent.

Proof (b) To say that T_A is onto is equivalent to saying that the system $A\mathbf{x} = \mathbf{b}$ has a solution for every vector $\mathbf{b}$ in R^m. But this is so if and only if every vector $\mathbf{b}$ in R^m is in the column space of A (Theorem 4.8.1), which is so if and only if the columns of A span R^m. ■

We leave it as an exercise to show that parts (*t*), (*u*), and (*v*) below can be added to Equivalence Theorem 8.2.4 in the case where $T_A : R^n \rightarrow R^n$ is a linear operator.

Theorem 8.2.4

Equivalent Statements

If A is an $n \times n$ matrix in which there are no duplicate rows and no duplicate columns, then the following statements are equivalent.

(*a*) A is invertible.

(*b*) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

(*c*) The reduced row echelon form of A is I_n.

(*d*) A is expressible as a product of elementary matrices.

(*e*) $A\mathbf{x} = \mathbf{b}$ is consistent for every $n \times 1$ matrix $\mathbf{b}$.

(*f*) $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $n \times 1$ matrix $\mathbf{b}$.

(*g*) $\det(A) \neq 0$.

(*h*) The column vectors of A are linearly independent.

(*i*) The row vectors of A are linearly independent.

(*j*) The column vectors of A span R^n.

(*k*) The row vectors of A span R^n.

(*l*) The column vectors of A form a basis for R^n.

(*m*) The row vectors of A form a basis for R^n.

(*n*) A has rank n.

(*o*) A has nullity 0.

(*p*) The orthogonal complement of the null space of A is R^n.

(*q*) The orthogonal complement of the row space of A is $\{\mathbf{0}\}$.

(*r*) $\lambda = 0$ is not an eigenvalue of A.

(*s*) A^TA is invertible.

(*t*) The kernel of T_A is $\{\mathbf{0}\}$.

(*u*) The range of T_A is R^n.

(*v*) T_A is one-to-one.

The key to solving a mathematical problem is often adopting the right point of view; and this is why, in linear algebra, we develop different ways of thinking about the same vector space. For example, if A is an $m \times n$ matrix, here are three ways of viewing the same subspace of R^n:

- **Matrix view:** the null space of A
- **System view:** the solution space of $A\mathbf{x} = \mathbf{0}$
- **Transformation view:** the kernel of T_A

and here are three ways of viewing the same subspace of R^m:

- **Matrix view:** the column space of A
- **System view:** all $\mathbf{b}$ in R^m for which $A\mathbf{x} = \mathbf{b}$ is consistent
- **Transformation view:** the range of T_A

Inverse Linear Transformations

In Section 1.9 we introduced the concept of an invertible matrix operator, and in this subsection we will extend that idea to general linear transformations. By way of review, recall that a matrix operator $T_A : R^n \rightarrow R^n$ is invertible if and only if the matrix A is invertible, in which case the inverse of that operator is $T_{A^{-1}} : R^n \rightarrow R^n$. In words, *the inverse of multiplication by A is multiplication by* A^{-1}.

EXAMPLE 8 | A One-to-One Matrix Transformation

Let $T : R^3 \rightarrow R^3$ be the linear operator defined by the formula

$$T(x_1, x_2, x_3) = (3x_1 + x_2, -2x_1 - 4x_2 + 3x_3, 5x_1 + 4x_2 - 2x_3)$$

Determine whether T is one-to-one; if so, find $T^{-1}(x_1, x_2, x_3)$.

Solution The stated formula defines a matrix transformation whose standard matrix by Formula (15) of Section 1.8 is

$$A = \begin{bmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{bmatrix}$$

(verify). This matrix is invertible and its inverse is

$$A^{-1} = \begin{bmatrix} 4 & -2 & -3 \\ -11 & 6 & 9 \\ -12 & 7 & 10 \end{bmatrix}$$

Thus, the transformation T is invertible and

$$T^{-1}\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = A^{-1}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 & -2 & -3 \\ -11 & 6 & 9 \\ -12 & 7 & 10 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4x_1 - 2x_2 - 3x_3 \\ -11x_1 + 6x_2 + 9x_3 \\ -12x_1 + 7x_2 + 10x_3 \end{bmatrix}$$

Expressing this result in comma delimited notation yields

$$T^{-1}(x_1, x_2, x_3) = (4x_1 - 2x_2 - 3x_3, -11x_1 + 6x_2 + 9x_3, -12x_1 + 7x_2 + 10x_3)$$

Now let us turn our attention to the invertibility of general linear transformations. If $T : V \to W$ is a one-to-one linear transformation with range $R(T)$, and if $\mathbf{w}$ is any vector in $R(T)$, then the fact that T is one-to-one means that there is *exactly one* vector $\mathbf{v}$ in V for which $T(\mathbf{v}) = \mathbf{w}$. This fact allows us to define a new function, called the ***inverse of T*** (and denoted by T^{-1}), that is defined on the range of T and that maps $\mathbf{w}$ back into $\mathbf{v}$ (**Figure 8.2.4**).

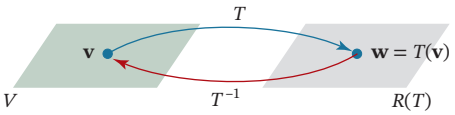


FIGURE 8.2.4 The inverse of T maps $T(\mathbf{v})$ back into $\mathbf{v}$.

In the exercises we will ask you to prove that $T^{-1} : R(T) \to V$ is a linear transformation. Moreover, it follows from the definition of T^{-1} that

$$\boxed{T^{-1}(T(\mathbf{v})) = T^{-1}(\mathbf{w}) = \mathbf{v}} \tag{1}$$

$$\boxed{T(T^{-1}(\mathbf{w})) = T(\mathbf{v}) = \mathbf{w}} \tag{2}$$

so that T and T^{-1}, when applied in succession in either order, cancel the effect of each other.

EXAMPLE 9 | An Inverse Transformation

We showed in Example 4 of this section that the linear transformation $T : P_n \to P_{n+1}$ given by

$$T(\mathbf{p}) = T(p(x)) = xp(x)$$

is one-to-one but not onto. The fact that it is not onto can be seen explicitly from the formula

$$T(c_0 + c_1x + \cdots + c_nx^n) = c_0x + c_1x^2 + \cdots + c_nx^{n+1} \tag{3}$$

The fact that T is not onto does not preclude the existence of an inverse, since the inverse is defined on the range of T. It is evident from (3) the range in this case consists of all polynomials of degree $n + 1$ or less that have a zero constant term and that the inverse is given by the formula

$$T^{-1}(c_0x + c_1x^2 + \cdots + c_nx^{n+1}) = c_0 + c_1x + \cdots + c_nx^n$$

For example, in the case where $n \geq 3$,

$$T^{-1}(2x - x^2 + 5x^3 + 3x^4) = 2 - x + 5x^2 + 3x^3$$

Composition of Linear Transformations

The following definition extends Formula (1) of Section 1.9 to general linear transformations.

Definition 3

If $T_1 : U \to V$ and $T_2 : V \to W$ are linear transformations, then the ***composition*** of T_2 with T_1, denoted by $T_2 \circ T_1$ (and which is read "T_2 circle T_1"), is the function defined by the formula

$$\boxed{(T_2 \circ T_1)(\mathbf{u}) = T_2(T_1(\mathbf{u}))} \tag{4}$$

where $\mathbf{u}$ is a vector in U.

Note that the word "with" establishes the order of the operations in a composition. The composition of T_2 *with* T_1 is

$$(T_2 \circ T_1)(\mathbf{u}) = T_2(T_1(\mathbf{u}))$$

whereas the composition of T_1 *with* T_2 is

$$(T_1 \circ T_2)(\mathbf{u}) = T_1(T_2(\mathbf{u}))$$

It is not true, in general, that $T_1 \circ T_2 = T_2 \circ T_1$.

Remark Observe that this definition requires that the domain of T_2 (which is V) contain the range of T_1. This is essential for the formula $T_2(T_1(\mathbf{u}))$ to make sense (**Figure 8.2.5**).

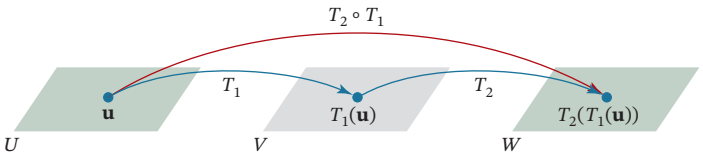


FIGURE 8.2.5 The composition of T_2 with T_1.

Our next theorem shows that the composition of two linear transformations is itself a linear transformation.

Theorem 8.2.5

If $T_1 : U \to V$ and $T_2 : V \to W$ are linear transformations, then $(T_2 \circ T_1) : U \to W$ is also a linear transformation.

Proof If $\mathbf{u}$ and $\mathbf{v}$ are vectors in U and c is a scalar, then it follows from (4) and the linearity of T_1 and T_2 that

$$\begin{aligned}(T_2 \circ T_1)(\mathbf{u}+\mathbf{v}) &= T_2(T_1(\mathbf{u}+\mathbf{v})) = T_2(T_1(\mathbf{u}) + T_1(\mathbf{v})) \\ &= T_2(T_1(\mathbf{u})) + T_2(T_1(\mathbf{v})) \\ &= (T_2 \circ T_1)(\mathbf{u}) + (T_2 \circ T_1)(\mathbf{v})\end{aligned}$$

and

$$\begin{aligned}(T_2 \circ T_1)(c\mathbf{u}) &= T_2(T_1(c\mathbf{u})) = T_2(cT_1(\mathbf{u})) \\ &= cT_2(T_1(\mathbf{u})) = c(T_2 \circ T_1)(\mathbf{u})\end{aligned}$$

Thus, $T_2 \circ T_1$ satisfies the two requirements of a linear transformation. ■

EXAMPLE 10 | Composition of Linear Transformations

Let $T_1 : P_1 \to P_2$ and $T_2 : P_2 \to P_2$ be the linear transformations given by the formulas

$$T_1(p(x)) = xp(x) \quad \text{and} \quad T_2(p(x)) = p(2x+4)$$

Then the composition $(T_2 \circ T_1) : P_1 \to P_2$ is given by the formula

$$(T_2 \circ T_1)(p(x)) = T_2(T_1(p(x))) = T_2(xp(x)) = (2x+4)p(2x+4)$$

In particular, if $p(x) = c_0 + c_1x$, then

$$\begin{aligned}(T_2 \circ T_1)(p(x)) &= (T_2 \circ T_1)(c_0 + c_1x) = (2x+4)(c_0 + c_1(2x+4)) \\ &= c_0(2x+4) + c_1(2x+4)^2\end{aligned}$$

EXAMPLE 11 | Composition with the Identity Operator

If $T : V \to V$ is any linear operator, and if $I : V \to V$ is the identity operator (Example 3 of Section 8.1), then for all vectors $\mathbf{v}$ in V, we have

$$(T \circ I)(\mathbf{v}) = T(I(\mathbf{v})) = T(\mathbf{v})$$
$$(I \circ T)(\mathbf{v}) = I(T(\mathbf{v})) = T(\mathbf{v})$$

It follows that $T \circ I$ and $I \circ T$ are the same as T; that is,

$$\boxed{T \circ I = T} \quad \text{and} \quad \boxed{I \circ T = T} \tag{5}$$

As illustrated in **Figure 8.2.6**, compositions can be defined for more than two linear transformations. For example, if

$$T_1 : U \to V, \quad T_2 : V \to W, \quad \text{and} \quad T_3 : W \to Y$$

are linear transformations, then the composition $T_3 \circ T_2 \circ T_1$ is defined by

$$\boxed{(T_3 \circ T_2 \circ T_1)(\mathbf{u}) = T_3(T_2(T_1(\mathbf{u})))} \tag{6}$$

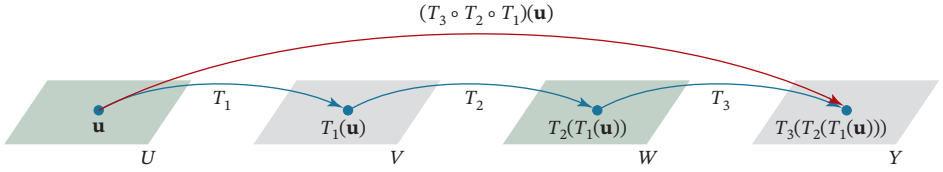


FIGURE 8.2.6 The composition of three linear transformations.

Composition of One-to-One Linear Transformations

Our next theorem shows that the composition of one-to-one linear transformations is one-to-one and that the inverse of a composition is the composition of the inverses in the reverse order.

Theorem 8.2.6

If $T_1 : U \to V$ and $T_2 : V \to W$ are one-to-one linear transformations, then:

(*a*) $T_2 \circ T_1$ is one-to-one.

(*b*) $(T_2 \circ T_1)^{-1} = T_1^{-1} \circ T_2^{-1}$.

Note the order of the subscripts on the two sides of the formula in part (*b*) of Theorem 8.2.5.

Proof (a) We want to show that $T_2 \circ T_1$ maps distinct vectors in U into distinct vectors in W. But if $\mathbf{u}$ and $\mathbf{v}$ are distinct vectors in U, then $T_1(\mathbf{u})$ and $T_1(\mathbf{v})$ are distinct vectors in V since T_1 is one-to-one. This and the fact that T_2 is one-to-one imply that

$$T_2(T_1(\mathbf{u})) \quad \text{and} \quad T_2(T_1(\mathbf{v}))$$

are also distinct vectors. But these expressions can also be written as

$$(T_2 \circ T_1)(\mathbf{u}) \quad \text{and} \quad (T_2 \circ T_1)(\mathbf{v})$$

so $T_2 \circ T_1$ maps $\mathbf{u}$ and $\mathbf{v}$ into distinct vectors in W.

Proof (b) We want to show that

$$(T_2 \circ T_1)^{-1}(\mathbf{w}) = (T_1^{-1} \circ T_2^{-1})(\mathbf{w})$$

for every vector $\mathbf{w}$ in the range of $T_2 \circ T_1$. For this purpose, let

$$\mathbf{u} = (T_2 \circ T_1)^{-1}(\mathbf{w}) \tag{7}$$

so our goal is to show that

$$\mathbf{u} = (T_1^{-1} \circ T_2^{-1})(\mathbf{w})$$

But it follows from (7) that

$$(T_2 \circ T_1)(\mathbf{u}) = \mathbf{w}$$

or, equivalently,

$$T_2(T_1(\mathbf{u})) = \mathbf{w}$$

Now, taking T_2^{-1} of each side of this equation, then taking T_1^{-1} of each side of the result, and then using (1) yields (verify)

$$\mathbf{u} = T_1^{-1}(T_2^{-1}(\mathbf{w}))$$

or, equivalently,

$$\mathbf{u} = (T_1^{-1} \circ T_2^{-1})(\mathbf{w}) \quad \blacksquare$$

In words, part (*b*) of Theorem 8.2.5 states that *the inverse of a composition is the composition of the inverses in the reverse order*. This result can be extended to compositions of three or more linear transformations; for example,

$$\boxed{(T_3 \circ T_2 \circ T_1)^{-1} = T_1^{-1} \circ T_2^{-1} \circ T_3^{-1}} \tag{8}$$

Part (*b*) of Theorem 8.2.5 and Formula (8) apply to general linear transformations. In the special case where they are matrix transformations they can be written as

$$(T_B \circ T_A)^{-1} = T_A^{-1} \circ T_B^{-1} \quad \text{and} \quad (T_C \circ T_B \circ T_A)^{-1} = T_A^{-1} \circ T_B^{-1} \circ T_C^{-1}$$

or equivalently as

$$\boxed{(T_{BA})^{-1} = T_{A^{-1}B^{-1}} \quad \text{and} \quad (T_{CBA})^{-1} = T_{A^{-1}B^{-1}C^{-1}}} \tag{9}$$

Exercise Set 8.2

In Exercises 1–2, determine whether the stated matrix operator is one-to-one.

1. **a.** The orthogonal projection onto the x-axis in R^2.

b. The reflection about the y-axis in R^2.

c. The reflection about the line $y = x$ in R^2.

2. **a.** A rotation about the z-axis in R^3.

b. A reflection about the xy-plane in R^3.

c. An orthogonal projection onto the xz-plane in R^3.

In Exercises 3–4, determine whether the linear transformation is one-to-one by finding its kernel and then applying Theorem 8.2.1.

3. **a.** $T : R^2 \to R^2$, where $T(x, y) = (y, x)$

b. $T : R^2 \to R^3$, where $T(x, y) = (x, y, x + y)$

c. $T : R^3 \to R^2$, where $T(x, y, z) = (x + y + z, x - y - z)$

4. **a.** $T : R^2 \to R^3$, where $T(x, y) = (x - y, y - x, 2x - 2y)$

b. $T : R^2 \to R^2$, where $T(x, y) = (0, 2x + 3y)$

c. $T : R^2 \to R^2$, where $T(x, y) = (x + y, x - y)$

In Exercises 5–6, determine whether multiplication by A is one-to-one by computing the nullity of A and then applying Theorem 8.2.1.

5. **a.** $A = \begin{bmatrix} 1 & -2 \\ 2 & -4 \\ -3 & 6 \end{bmatrix}$

b. $A = \begin{bmatrix} 1 & 3 & 1 & 7 \\ 2 & 7 & 2 & 4 \\ -1 & -3 & 0 & 0 \end{bmatrix}$

6. **a.** $A = \begin{bmatrix} 1 & 2 \\ 2 & 7 \\ 3 & 9 \end{bmatrix}$

b. $A = \begin{bmatrix} 1 & -3 & 6 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

7. Use the given information to determine whether the linear transformation is one-to-one.

a. $T : V \to W$; nullity$(T) = 0$

b. $T : V \to W$; rank$(T) = \dim(V)$

c. $T : V \to W$; $\dim(W) < \dim(V)$

8. Use the given information to determine whether the linear operator is one-to-one, onto, both, or neither.

a. $T : V \to V$; nullity$(T) = 0$

b. $T : V \to V$; rank$(T) < \dim(V)$

c. $T : V \to V$; $R(T) = V$

9. Show that the linear transformation $T : P_2 \to R^2$ defined by $T(p(x)) = (p(-1), p(1))$ is not one-to-one by finding a nonzero polynomial that maps into $\mathbf{0} = (0, 0)$. Do you think that this transformation is onto?

10. Show that the linear transformation $T : P_2 \to P_2$ defined by $T(p(x)) = p(x + 1)$ is one-to-one. Do you think that this transformation is onto?

11. Let $\mathbf{a}$ be a fixed vector in R^3. Does the formula $T(\mathbf{v}) = \mathbf{a} \times \mathbf{v}$ define a one-to-one linear operator on R^3? Explain your reasoning.

12. Let E be a fixed 2×2 elementary matrix. Does the formula $T(A) = EA$ define a one-to-one linear operator on M_{22}? Explain your reasoning.

In Exercises 13–14, use Theorem 8.2.3 to determine whether multiplication by A is one-to-one, onto, both, or neither. Justify your answer.

13. a. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 5 \end{bmatrix}$ **b.** $A = \begin{bmatrix} 1 & 3 & 1 & 1 \\ 1 & 3 & 1 & 0 \\ 1 & 4 & 0 & 0 \end{bmatrix}$

c. $A = \begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix}$ **d.** $A = \begin{bmatrix} -2 & 1 & 0 \\ 6 & -3 & 1 \\ 8 & -4 & 3 \end{bmatrix}$

14. a. $A = \begin{bmatrix} 9 & -3 \\ -4 & 2 \\ 1 & 1 \end{bmatrix}$ **b.** $A = \begin{bmatrix} 3 & -3 & 1 & 1 \\ 6 & -6 & 0 & 2 \\ 9 & -9 & 1 & 3 \end{bmatrix}$

c. $A = \begin{bmatrix} 3 & -9 \\ -1 & 3 \end{bmatrix}$ **d.** $A = \begin{bmatrix} 2 & 3 & 8 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$

In Exercises 15–16, describe in words the inverse of the given one-to-one operator.

15. a. The reflection about the x-axis on R^2.

b. The rotation about the origin through an angle of $\pi/4$ on R^2.

16. a. The reflection about the yz-plane in R^3.

b. The rotation through an angle of $-18°$ about the z-axis in R^3.

In Exercises 17–18, use matrix inversion to confirm the stated result in R^2.

17. a. The inverse transformation for a reflection about $y = x$ is a reflection about $y = x$.

b. The inverse transformation for a rotation about the origin is a rotation about the origin.

18. a. The inverse transformation for a reflections about a coordinate axis is a reflection about that axis.

b. The inverse transformation for a reflection about the origin is a reflection about the origin.

19. Let $T : P_1 \to R^2$ be the function defined by the formula

$$T(p(x)) = (p(0), p(1))$$

a. Find $T(1 - 2x)$.

b. Show that T is a linear transformation.

c. Show that T is one-to-one.

d. Find $T^{-1}(2, 3)$, and sketch its graph.

20. In each part, determine whether $T : R^n \to R^n$ is one-to-one; if so, find $T^{-1}(x_1, x_2, \ldots, x_n)$.

a. $T(x_1, x_2, \ldots, x_n) = (0, x_1, x_2, \ldots, x_{n-1})$

b. $T(x_1, x_2, \ldots, x_n) = (x_n, x_{n-1}, \ldots, x_2, x_1)$

c. $T(x_1, x_2, \ldots, x_n) = (x_2, x_3, \ldots, x_n, x_1)$

21. Let $T : R^n \to R^n$ be the linear operator defined by the formula

$$T(x_1, x_2, \ldots, x_n) = (a_1 x_1, a_2 x_2, \ldots, a_n x_n)$$

where $a_1, \ldots, a_n$ are constants.

a. Under what conditions will T have an inverse?

b. Assuming that the conditions determined in part (a) are satisfied, find a formula for $T^{-1}(x_1, x_2, \ldots, x_n)$.

22. Let $T_A : R^4 \to R^2$ be multiplication by the matrix

$$A = \begin{bmatrix} 1 & 0 & 2 & 5 \\ 3 & 4 & 1 & 3 \end{bmatrix}$$

Find parametric equations for the set of vectors that map into the vector $(1, 1)$, if any.

In Exercises 23–24, compute $(T_2 \circ T_1)(x, y)$.

23. $T_1(x, y) = (2x, 3y), T_2(x, y) = (x - y, x + y)$

24. $T_1(x, y) = (2x, -3y, x + y), T_2(x, y, z) = (x - y, y + z)$

25. Suppose that the linear transformations $T_1 : P_2 \to P_2$ and $T_2 : P_2 \to P_3$ are given by the formulas $T_1(p(x)) = p(x + 1)$ and $T_2(p(x)) = xp(x)$. Find $(T_2 \circ T_1)(a_0 + a_1 x + a_2 x^2)$.

26. Let $T_1 : P_n \to P_n$ and $T_2 : P_n \to P_n$ be the linear operators given by $T_1(p(x)) = p(x - 1)$ and $T_2(p(x)) = p(x + 1)$. Find $(T_1 \circ T_2)(p(x))$ and $(T_2 \circ T_1)(p(x))$.

27. Let $T_1 : M_{22} \to R$ and $T_2 : M_{22} \to M_{22}$ be the linear transformations given by $T_1(A) = \text{tr}(A)$ and $T_2(A) = A^T$.

a. Find $(T_1 \circ T_2)(A)$, where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

b. Can you find $(T_2 \circ T_1)(A)$? Explain.

28. Rework Exercise 27 if $T_1 : M_{22} \to M_{22}$ and $T_2 : M_{22} \to M_{22}$ are the linear transformations, $T_1(A) = kA$ and $T_2(A) = A^T$, where k is a scalar.

In Exercises 29–30, compute $(T_3 \circ T_2 \circ T_1)(x, y)$.

29. $T_1(x, y) = (-2y, 3x, x - 2y), T_2(x, y, z) = (y, z, x)$, $T_3(x, y, z) = (x + z, y - z)$

30. $T_1(x,y) = (x+y, y, -x)$, $T_2(x,y,z) = (0, x+y+z, 3y)$, $T_3(x,y,z) = (3x+2y, 4z-x-3y)$

31. Let $T_1 : P_2 \to P_3$ and $T_2 : P_3 \to P_3$ be the linear transformations given by the formulas

$$T_1(p(x)) = xp(x) \quad \text{and} \quad T_2(p(x)) = p(x+1)$$

a. Find formulas for

$$T_1^{-1}(p(x)), T_2^{-1}(p(x))$$

and

$$(T_1^{-1} \circ T_2^{-1})(p(x))$$

b. Verify that $(T_2 \circ T_1)^{-1} = T_1^{-1} \circ T_2^{-1}$.

32. Let $T_1 : R^2 \to R^2$ and $T_2 : R^2 \to R^2$ be the linear operators given by the formulas

$$T_1(x,y) = (x+y, x-y) \quad \text{and} \quad T_2(x,y) = (2x+y, x-2y)$$

a. Show that T_1 and T_2 are one-to-one.

b. Find formulas for

$$T_1^{-1}(x,y), \quad T_2^{-1}(x,y), \quad (T_2 \circ T_1)^{-1}(x,y)$$

c. Verify that $(T_2 \circ T_1)^{-1} = T_1^{-1} \circ T_2^{-1}$.

33. Let $T_1 : V \to V$ be the linear operator given by $T_1(\mathbf{v}) = 4\mathbf{v}$. Find a linear operator $T_2 : V \to V$ such that $T_1 \circ T_2 = I$ and $T_2 \circ T_1 = I$.

34. Let $T_1 : M_{22} \to P_1$ and $T_2 : P_1 \to R^3$ be the linear transformations given by

$$T_1\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + (c+d)x$$

and $T_2(a+bx) = (a, b, a)$.

a. Find the formula for $T_2 \circ T_1$.

b. Show that $T_2 \circ T_1$ is not one-to-one by finding distinct 2×2 matrices A and B such that

$$(T_2 \circ T_1)(A) = (T_2 \circ T_1)(B)$$

c. Show that $T_2 \circ T_1$ is not onto by finding a vector (a, b, c) in R^3 that is not in the range of $T_2 \circ T_1$.

35. Let $T : R^3 \to R^3$ be the orthogonal projection of R^3 onto the xy-plane. Show that $T \circ T = T$.

36. ***(Calculus required)*** Let V be the vector space $C^1[0,1]$ and let $T : V \to R$ be defined by

$$T(\mathbf{f}) = f(0) + 2f'(0) + 3f'(1)$$

Verify that T is a linear transformation. Determine whether T is one-to-one, and justify your conclusion.

37. ***(Calculus required)*** The Fundamental Theorem of Calculus implies that integration and differentiation reverse the actions of each other. Define a transformation $D : P_n \to P_{n-1}$ by $D(p(x)) = p'(x)$, and define $J : P_{n-1} \to P_n$ by

$$J(p(x)) = \int_0^x p(t)\,dt$$

a. Show that D and J are linear transformations.

b. Explain why J is not the inverse transformation of D.

c. Can the domains and/or codomains of D and J be restricted so they are inverse linear transformations?

38. ***(Calculus required)*** Let

$$D(\mathbf{f}) = f'(x) \quad \text{and} \quad J(\mathbf{f}) = \int_0^x f(t)\,dt$$

be the linear transformations in Examples 11 and 12 of Section 8.1. Find $(J \circ D)(\mathbf{f})$ for

a. $\mathbf{f}(x) = x^2 + 3x + 2$. **b.** $\mathbf{f}(x) = \sin x$.

39. ***(Calculus required)*** Let $J : P_1 \to R$ be the integration transformation $J(\mathbf{p}) = \int_{-1}^{1} p(x)dx$. Determine whether J is one-to-one. Justify your answer.

40. ***(Calculus required)*** Let $D : P_n \to P_{n-1}$ be the differentiation transformation $D(p(x)) = p'(x)$. Determine whether D is onto, and justify your answer.

41. Let A be an $n \times n$ matrix such that $\det(A) = 0$, and let $T : R^n \to R^n$ be multiplication by A.

a. What can you say about the range of the matrix operator T? Give an example that illustrates your conclusion.

b. What can you say about the number of vectors that T maps into $\mathbf{0}$?

42. Answer the questions in Exercise 41 in the case where $\det(A) \neq 0$.

43. **a.** Is a composition of one-to-one matrix transformations one-to-one? Justify your conclusion.

b. Can the composition of a one-to-one matrix transformation and a matrix transformation that is not one-to-one be one-to-one? Account for both possible orders of composition and justify your conclusion.

Working with Proofs

44. Prove: If there exists an onto linear transformation $T : V \to W$ then $\dim(V) \geq \dim(W)$.

45. Prove: If $T : V \to W$ is a one-to-one linear transformation, then $T^{-1} : R(T) \to V$ is a one-to-one linear transformation.

46. Use the definition of $T_3 \circ T_2 \circ T_1$ given by Formula (6) to prove that

a. $T_3 \circ T_2 \circ T_1$ is a linear transformation.

b. $T_3 \circ T_2 \circ T_1 = (T_3 \circ T_2) \circ T_1$.

c. $T_3 \circ T_2 \circ T_1 = T_3 \circ (T_2 \circ T_1)$.

47. Let V and W be finite-dimensional vector space and let $T : V \to W$ be a linear transformation. Prove:

a. If $\dim(V) > \dim(W)$, then T cannot be one-to-one.

b. If $\dim(V) < \dim(W)$, then T cannot be onto.

48. Add parts (t), (u), and (v) to Equivalence Theorem 8.2.4 by proving that each of those statements is equivalent to the invertibility of A.

True-False Exercises

TF. In parts **(a)–(j)** determine whether the statement is true or false, and justify your answer.

a. $T : V \to W$ is one-to-one if and only if $T(\mathbf{u}) \neq T(\mathbf{v})$ whenever $\mathbf{u} \neq \mathbf{v}$.

b. $T : V \to W$ is one-to-one if and only if for each vector $\mathbf{w}$ in the range of T there is exactly one vector $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{w}$.

c. The inverse of a one-to-one linear transformation is a linear transformation.

d. If a linear transformation T has an inverse, then the kernel of T is the zero subspace.

e. If $T : R^2 \to R^2$ is the orthogonal projection onto the x-axis, then $T^{-1} : R^2 \to R^2$ maps each point on the x-axis onto a line that is perpendicular to the x-axis.

f. If $T_1 : U \to V$ and $T_2 : V \to W$ are linear transformations, and if T_1 is not one-to-one, then neither is $T_2 \circ T_1$.

g. If A is an $n \times n$ matrix and if the linear system $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then the range of the matrix operator is not R^n.

h. If T_A and T_B are matrix operators on R^n, then $T_A(T_B(\mathbf{x})) = T_B(T_A(\mathbf{x}))$ for every vector $\mathbf{x}$ in R^n.

i. The kernel of a matrix transformation $T_A : R^n \to R^m$ is the same as the null space of A.

j. If there is a nonzero vector in the kernel of the matrix operator $T_A : R^n \to R^n$, then this operator is not one-to-one.

Working with Technology

T1. Consider the matrix transformation $T_A : R^3 \to R^4$, where

$$A = \begin{bmatrix} .23 & -.02 & .67 \\ 1.12 & .10 & .44 \\ -.03 & .11 & .12 \\ .09 & -.68 & .83 \end{bmatrix}$$

Use Theorem 8.2.3 to determine whether T_A is one-to-one.

T2. Consider the matrix transformation $T_B : R^4 \to R^3$, where

$$B = \begin{bmatrix} .52 & .42 & .91 & -.05 \\ -.01 & 1.11 & .37 & .78 \\ .21 & .73 & -.32 & .24 \end{bmatrix}$$

Use Theorem 8.2.3 to determine whether T_B is onto.

8.3 Isomorphism

In this section we will establish a fundamental connection between real finite-dimensional vector spaces and the Euclidean space R^n. This connection is not only important theoretically, but it has practical applications in that is allows us to perform vector computations in certain general vector spaces by working with the vectors in R^n.

Isomorphism

Although many of the theorems in this text have been concerned exclusively with the vector space R^n, this is not as limiting as it might seem. We will show that the vector space R^n is the "mother" of all real n-dimensional vector spaces in the sense that every n-dimensional vector space must have the same algebraic structure as R^n even though its vectors may not be expressed as n-tuples. To explain what we mean by this, we will need the following definition.

Definition 1

A linear transformation $T : V \to W$ that is both one-to-one and onto is said to be an ***isomorphism***, and W is said to be ***isomorphic*** to V.

In the exercises we will ask you to show that if $T : V \to W$ is an isomorphism, then $T^{-1} : W \to V$ is also an isomorphism. Accordingly, we will usually say simply that ***V and W are isomorphic*** and that ***T is an isomorphism between V and W***.

The word *isomorphic* is derived from the Greek words *iso*, meaning "identical," and *morphe*, meaning "form." This terminology is appropriate because, as we will now explain, isomorphic vector spaces have the same "algebraic form," even though they may consist of

different kinds of objects. For example, the following diagram illustrates an isomorphism between P_2 and R^3

$$c_0 + c_1x + c_2x^2 \underset{T^{-1}}{\overset{T}{\rightleftarrows}} (c_0, c_1, c_2)$$

Although the vectors on the two sides of the arrows are different kinds of objects, the vector operations on each side mirror those on the other side. For example, for scalar multiplication we have

$$k(c_0 + c_1x + c_2x^2) \underset{T^{-1}}{\overset{T}{\rightleftarrows}} k(c_0, c_1, c_2)$$

$$kc_0 + kc_1x + kc_2x^2 \underset{T^{-1}}{\overset{T}{\rightleftarrows}} (kc_0, kc_1, kc_2)$$

and for vector addition we have

$$(c_0 + c_1x + c_2x^2) + (d_0 + d_1x + d_2x^2) \underset{T^{-1}}{\overset{T}{\rightleftarrows}} (c_0, c_1, c_2) + (d_0, d_1, d_2)$$

$$(c_0 + d_0) + (c_1 + d_1)x + (c_2 + d_2)x^2 \underset{T^{-1}}{\overset{T}{\rightleftarrows}} (c_0 + d_0, c_1 + d_1, c_2 + d_2)$$

The following theorem, which is one of the most basic results in linear algebra, reveals the fundamental importance of the vector space R^n.

Theorem 8.3.1

Every real n-dimensional vector space is isomorphic to R^n.

Theorem 8.3.1 tells us that every real n-dimensional vector space differs from R^n only in notation; the algebraic structures of the two spaces are the same.

Proof Let V be a real n-dimensional vector space. To prove that V is isomorphic to R^n we must find a linear transformation $T : V \to R^n$ that is one-to-one and onto. For this purpose, let

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$$

be any basis for V, let

$$\mathbf{u} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \cdots + k_n\mathbf{v}_n \tag{1}$$

be the representation of a vector $\mathbf{u}$ in V as a linear combination of the basis vectors, and let $T : V \to R^n$ be the coordinate map

$$T(\mathbf{u}) = (\mathbf{u})_S = (k_1, k_2, \ldots, k_n) \tag{2}$$

We will show that T is linear, one-to-one, and onto and hence is an isomorphism. To prove the linearity, let $\mathbf{u}$ and $\mathbf{v}$ be vectors in V, let c be a scalar, and let

$$\mathbf{u} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \cdots + k_n\mathbf{v}_n \quad \text{and} \quad \mathbf{v} = d_1\mathbf{v}_1 + d_2\mathbf{v}_2 + \cdots + d_n\mathbf{v}_n \tag{3}$$

be the representations of $\mathbf{u}$ and $\mathbf{v}$ as linear combinations of the basis vectors. Then it follows from (3) that

$$\begin{aligned} T(c\mathbf{u}) &= T(ck_1\mathbf{v}_1 + ck_2\mathbf{v}_2 + \cdots + ck_n\mathbf{v}_n) \\ &= (ck_1, ck_2, \ldots, ck_n) \\ &= c(k_1, k_2, \ldots, k_n) = cT(\mathbf{u}) \end{aligned}$$

and that

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T\big((k_1 + d_1)\mathbf{v}_1 + (k_2 + d_2)\mathbf{v}_2 + \cdots + (k_n + d_n)\mathbf{v}_n\big) \\ &= (k_1 + d_1, k_2 + d_2, \ldots, k_n + d_n) \\ &= (k_1, k_2, \ldots, k_n) + (d_1, d_2, \ldots, d_n) \\ &= T(\mathbf{u}) + T(\mathbf{v}) \end{aligned}$$

which shows that T is linear. To show that T is one-to-one, we must show that if $\mathbf{u}$ and $\mathbf{v}$ are distinct vectors in V, then so are their images in R^n. But if $\mathbf{u} \neq \mathbf{v}$, and if the representations of these vectors in terms of the basis vectors are as in (3), then we must have $k_i \neq d_i$ for at least one i. Thus,

$$T(\mathbf{u}) = (k_1, k_2, \ldots, k_n) \neq (d_1, d_2, \ldots, d_n) = T(\mathbf{v})$$

which shows that $\mathbf{u}$ and $\mathbf{v}$ have distinct images under T. Finally, the transformation T is onto, for if

$$\mathbf{w} = (k_1, k_2, \ldots, k_n)$$

is any vector in R^n, then it follows from (2) that $\mathbf{w}$ is the image under T of the vector

$$\mathbf{u} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \cdots + k_n\mathbf{v}_n \quad \blacksquare$$

Whereas Theorem 8.3.1 tells us, in general, that every real n-dimensional vector space is isomorphic to R^n, it is Formula (2) in its proof that tells us how to find isomorphisms.

Theorem 8.3.2

If S is an ordered basis for a vector space V, then the coordinate map

$$\mathbf{u} \xrightarrow{T} (\mathbf{u})_S$$

is an isomorphism between V and R^n.

Remark Recall that coordinate maps depend on the order in which the basis vectors are listed. Thus, Theorem 8.3.2 actually describes many possible isomorphisms, one for each of the $n!$ possible orders in which the basis vectors can be listed.

EXAMPLE 1 | The Natural Isomorphism Between P_{n-1} and R^n

It follows from Theorem 8.3.2 that the coordinate map

$$a_0 + a_1x + \cdots + a_{n-1}x^{n-1} \xrightarrow{T} (a_0, a_1, \ldots, a_{n-1})$$

defines an isomorphism between P_{n-1} and R^n. This is called the ***natural isomorphism*** between those vector spaces.

EXAMPLE 2 | The Natural Isomorphism Between M_{22} and R^4

It follows from Theorem 8.3.2 that the coordinate map

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{T} (a, b, c, d)$$

defines an isomorphism between M_{22} and R^4. This is a special case of the isomorphism that maps an $m \times n$ matrix into its coordinate vector. We call this the ***natural isomorphism*** between M_{mn} and R^{mn}.

CALCULUS REQUIRED

EXAMPLE 3 | Differentiation by Matrix Multiplication

Consider the differentiation transformation $D : P_3 \to P_2$ on the vector space of polynomials of degree 3 or less. If we map P_3 and P_2 into R^4 and R^3, respectively, by the natural isomorphisms, then the transformation D produces a corresponding matrix transformation from R^4 to R^3. Specifically, the derivative transformation

$$a_0 + a_1x + a_2x^2 + a_3x^3 \xrightarrow{D} a_1 + 2a_2x + 3a_3x^2$$

produces the matrix transformation

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} a_1 \\ 2a_2 \\ 3a_3 \end{bmatrix}$$

Thus, for example, the derivative

$$\frac{d}{dx}(2 + x + 4x^2 - x^3) = 1 + 8x - 3x^2$$

can be calculated as the matrix product

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 4 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ -3 \end{bmatrix}$$

This idea is useful for constructing numerical algorithms to calculate derivatives.

EXAMPLE 4 | Working with Isomorphisms

Use the natural isomorphism between P_5 and R^6 to determine whether the following polynomials are linearly independent.

$$\mathbf{p}_1 = 1 + 2x - 3x^2 + 4x^3 + x^5$$
$$\mathbf{p}_2 = 1 + 3x - 4x^2 + 6x^3 + 5x^4 + 4x^5$$
$$\mathbf{p}_3 = 3 + 8x - 11x^2 - 16x^3 + 10x^4 + 9x^5$$

Solution We will convert this to a matrix problem by creating a matrix whose rows are the coordinate vectors of the polynomials under the natural isomorphism and then determine whether those rows are linearly independent using elementary row operations.

The matrix whose rows are the coordinate vectors of the polynomials under the natural isomorphism is

$$A = \begin{bmatrix} 1 & 2 & -3 & 4 & 0 & 1 \\ 1 & 3 & -4 & 6 & 5 & 4 \\ 3 & 8 & -11 & 16 & 10 & 9 \end{bmatrix}$$

We leave it for you to use elementary row operations to reduce this matrix to the row echelon form

$$R = \begin{bmatrix} 1 & 2 & -3 & 4 & 0 & 1 \\ 0 & 1 & -1 & 2 & 5 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

This matrix has only two nonzero rows, so the row space of A is two-dimensional. This means that its row vectors are linearly dependent and hence so are the polynomials.

Inner Product Space Isomorphisms

In the case where V is a real n-dimensional inner product space, both V and R^n have, in addition to their algebraic structure, a geometric structure arising from their respective inner products. Thus, it is reasonable to inquire if there exists an isomorphism from V to R^n that preserves the geometric structure as well as the algebraic structure. For example, we would want orthogonal vectors in V to have orthogonal counterparts in R^n, and we would want orthonormal sets in V to correspond to orthonormal sets in R^n.

In order for an isomorphism to preserve geometric structure, it obviously has to preserve inner products, since notions of length, angle, and orthogonality are all based on the inner product. Thus, if V and W are inner product spaces, then we call an isomorphism $T : V \to W$ an ***inner product space isomorphism*** if

$$\langle T(\mathbf{u}), T(\mathbf{v})\rangle = \langle \mathbf{u}, \mathbf{v}\rangle \quad \text{for all } \mathbf{u} \text{ and } \mathbf{v} \text{ in } V$$

Remark Keep in mind that the inner product on the left side of this equation is for W and that on the right is for V.

The following analog of Theorem 8.3.2 provides an important method for obtaining inner product space isomorphisms between real inner product spaces and Euclidean vector spaces.

Theorem 8.3.3

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is an ordered orthonormal basis for a real n-dimensional inner product space V, then the coordinate map

$$\mathbf{u} \xrightarrow{T} (\mathbf{u})_S$$

is an inner product space isomorphism between V and the vector space R^n with the Euclidean inner product.

EXAMPLE 5 | An Inner Product Space Isomorphism

We saw in Example 1 that the coordinate map

$$a_0 + a_1x + \cdots + a_{n-1}x^{n-1} \xrightarrow{T} (a_0, a_1, \ldots, a_{n-1})$$

with respect to the standard basis for P_{n-1} is an isomorphism between P_{n-1} and R^n. However, the standard basis is orthonormal with respect to the standard inner product on P_{n-1} (see Example 3 of Section 6.3), so it follows that T is actually an *inner product space isomorphism* with respect to the standard inner product on P_{n-1} and the Euclidean inner product on R^n. To verify that this is so, recall from Example 7 of Section 6.1 that the standard inner product on P_{n-1} of two vectors

$$\mathbf{p} = a_0 + a_1x + \cdots + a_{n-1}x^{n-1} \quad \text{and} \quad \mathbf{q} = b_0 + b_1x + \cdots + b_{n-1}x^{n-1}$$

is

$$\langle \mathbf{p}, \mathbf{q}\rangle = a_0b_0 + a_1b_1 + \cdots + a_{n-1}b_{n-1}$$

But this is exactly the Euclidean inner product on R^n of the n-tuples

$$(a_0, a_1, \ldots, a_{n-1}) \quad \text{and} \quad (b_0, b_1, \ldots, b_{n-1})$$

EXAMPLE 6 | A Notational Matter

Let R^n be the vector space of real n-tuples in comma-delimited form, let M_n be the vector space of real $n \times 1$ matrices, let R^n have the Euclidean inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v}$, and let M_n have the inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T\mathbf{v}$ in which $\mathbf{u}$ and $\mathbf{v}$ are expressed in column form. The mapping $T : R^n \rightarrow M_n$ defined by

$$(v_1, v_2, \ldots, v_n) \xrightarrow{T} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

is an inner product space isomorphism, so the distinction between the inner product space R^n and the inner product space M_n is essentially notational, a fact that we have used many times in this text.

Exercise Set 8.3

In Exercises 1–8, state whether the transformation is an isomorphism. No proof required.

1. $c_0 + c_1x \rightarrow (c_0 - c_1, c_1)$ from P_1 to R^2.
2. $(x, y) \rightarrow (x, y, 0)$ from R^2 to R^3.
3. $a + bx + cx^2 + dx^3 \rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ from P_3 to M_{22}.
4. $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow ad - bc$ from M_{22} to R.
5. $(a, b, c, d) \rightarrow a + bx + cx^2 + (d + 1)x^3$ from R^4 to P_3.
6. $A \rightarrow A^T$ from M_{nn} to M_{nn}.
7. $c_1 \sin x + c_2 \cos x \rightarrow (c_1, c_2)$ from the subspace of $C(-\infty, \infty)$ spanned by $S = \{\sin x, \cos x\}$ to R^2.
8. The map $(u_1, u_2, \ldots, u_n, \ldots) \rightarrow (0, u_1, u_2, \ldots, u_n, \ldots)$ from R^∞ to R^∞.
9. **a.** Find an isomorphism between the vector space of all 3×3 symmetric matrices and R^6.
   b. Find two different isomorphisms between the vector space of all 2×2 matrices and R^4.
10. **a.** Find an isomorphism between the vector space of all polynomials of degree at most 3 such that $p(0) = 0$ and R^3.
    b. Find an isomorphism between the vector spaces $\text{span}\{1, \sin(x), \cos(x)\}$ and R^3.

In Exercises 11–12, determine whether the matrix transformation $T_A : R^3 \rightarrow R^3$ is an isomorphism.

11. $A = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 2 \\ -1 & 1 & 0 \end{bmatrix}$
12. $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 2 \\ -1 & 1 & 0 \end{bmatrix}$

In Exercises 13–14, find the dimension n of the solution space W of $A\mathbf{x} = \mathbf{0}$, and then construct an isomorphism between W and R^n.

13. $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \end{bmatrix}$
14. $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

In Exercises 15–16, determine whether the transformation is an isomorphism from M_{22} to R^4.

15. $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \begin{bmatrix} a \\ a + b \\ a + b + c \\ a + b + c + d \end{bmatrix}$
16. $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \begin{bmatrix} a + b \\ a + b \\ a + b + c \\ a + b + c + d \end{bmatrix}$
17. Do you think that R^2 is isomorphic to the xy-plane in R^3? Justify your answer.
18. **a.** For what value or values of k, if any, is M_{mn} isomorphic to R^k?
    b. For what value or values of k, if any, is M_{mn} isomorphic to P_k?
19. Let $T : P_2 \rightarrow M_{22}$ be the mapping
$$T(\mathbf{p}) = T(p(x)) = \begin{bmatrix} p(0) & p(1) \\ p(1) & p(0) \end{bmatrix}$$
Is this an isomorphism? Justify your answer.
20. Show that if M_{22} and P_3 have the standard inner products given in Examples 6 and 7 of Section 6.1, then the mapping
$$\begin{bmatrix} a_0 & a_1 \\ a_2 & a_3 \end{bmatrix} \rightarrow a_0 + a_1x + a_2x^2 + a_3x^3$$
is an inner product space isomorphism between those spaces.

21. **(*Calculus required*)** Devise a method for using matrix multiplication to differentiate functions in the vector space $\text{span}\{1, \sin(x), \cos(x), \sin(2x), \cos(2x)\}$. Use your method to find the derivative of $3 - 4\sin(x) + \sin(2x) + 5\cos(2x)$.

Working with Proofs

22. Prove that if $T : V \to W$ is an isomorphism, then so is $T^{-1} : W \to V$.

23. Prove that if U, V, and W are vector spaces such that U is isomorphic to V and V is isomorphic to W, then U is isomorphic to W.

24. Use the result in Exercise 22 to prove that any two real finite-dimensional vector spaces with the same dimension are isomorphic to one another.

25. Prove that an inner product space isomorphism preserves angles and distances—that is, the angle between $\mathbf{u}$ and $\mathbf{v}$ in V is equal to the angle between $T(\mathbf{u})$ and $T(\mathbf{v})$ in W, and $\|\mathbf{u} - \mathbf{v}\|_V = \|T(\mathbf{u}) - T(\mathbf{v})\|_W$.

26. Prove that an inner product space isomorphism maps orthonormal sets into orthonormal sets.

True-False Exercises

TF. In parts **(a)–(f)** determine whether the statement is true or false, and justify your answer.

a. The vector spaces R^2 and P_2 are isomorphic.

b. If the kernel of a linear transformation $T : P_3 \to P_3$ is $\{\mathbf{0}\}$, then T is an isomorphism.

c. Every linear transformation from M_{33} to P_9 is an isomorphism.

d. There is a subspace of M_{23} that is isomorphic to R^4.

e. Isomorphic finite-dimensional vector spaces must have the same number of basis vectors.

f. R^n is isomorphic to a subspace of R^{n+1}.

8.4 Matrices for General Linear Transformations

In this section we will show that a general linear transformation from any n-dimensional vector space V to any m-dimensional vector space W can be performed using an appropriate matrix transformation from R^n to R^m. This idea is used in computer computations since computers are well suited for performing matrix computations.

Matrices of Linear Transformations

Suppose that V is an n-dimensional vector space, that W is an m-dimensional vector space, and that $T : V \to W$ is a linear transformation. Suppose further that B is a basis for V, that B' is a basis for W, and that for each vector $\mathbf{x}$ in V, the coordinate vectors for $\mathbf{x}$ and $T(\mathbf{x})$ are $[\mathbf{x}]_B$ and $[T(\mathbf{x})]_{B'}$, respectively (**Figure 8.4.1**).

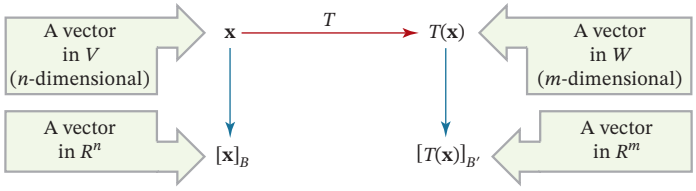


FIGURE 8.4.1

It will be our goal to find an $m \times n$ matrix A such that multiplication by A maps the vector $[\mathbf{x}]_B$ into the vector $[T(\mathbf{x})]_{B'}$ for each $\mathbf{x}$ in V (**Figure 8.4.2*a***). If we can do so, then, as illustrated in **Figure 8.4.2*b***, we will be able to execute the linear transformation T by using matrix multiplication and the following *indirect* procedure:

Finding $T(\mathbf{x})$ Indirectly

Step 1. Compute the coordinate vector $[\mathbf{x}]_B$.

Step 2. Multiply $[\mathbf{x}]_B$ on the left by A to produce $[T(\mathbf{x})]_{B'}$.

Step 3. Reconstruct $T(\mathbf{x})$ from its coordinate vector $[T(\mathbf{x})]_{B'}$.

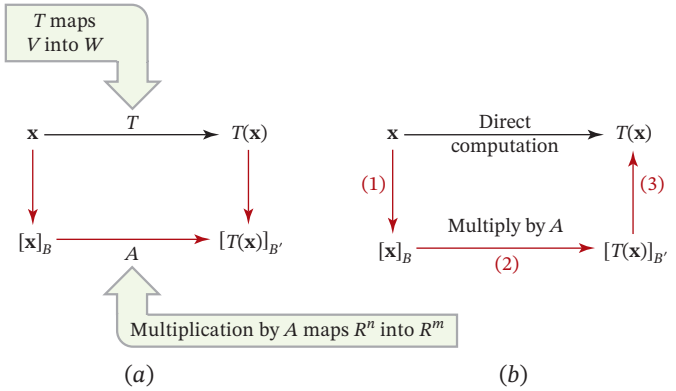


FIGURE 8.4.2

The key to executing this plan is to find an $m \times n$ matrix A with the property that

$$A[\mathbf{x}]_B = [T(\mathbf{x})]_{B'} \tag{1}$$

For this purpose, let $B = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be a basis for the n-dimensional space V and $B' = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\}$ a basis for the m-dimensional space W. Since Equation (1) must hold for all vectors in V, it must hold, in particular, for the basis vectors in B; that is,

$$A[\mathbf{u}_1]_B = [T(\mathbf{u}_1)]_{B'}, \quad A[\mathbf{u}_2]_B = [T(\mathbf{u}_2)]_{B'}, \ldots, \quad A[\mathbf{u}_n]_B = [T(\mathbf{u}_n)]_{B'} \tag{2}$$

But

$$[\mathbf{u}_1]_B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad [\mathbf{u}_2]_B = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \ldots, \quad [\mathbf{u}_n]_B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

so

$$A[\mathbf{u}_1]_B = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}$$

$$A[\mathbf{u}_2]_B = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}$$

$$\vdots \qquad\qquad \vdots \qquad\qquad \vdots$$

$$A[\mathbf{u}_n]_B = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

Substituting these results into (2) yields

$$\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} = [T(\mathbf{u}_1)]_{B'}, \quad \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} = [T(\mathbf{u}_2)]_{B'}, \ldots, \quad \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = [T(\mathbf{u}_n)]_{B'}$$

which shows that the successive columns of A are the coordinate vectors of

$$T(\mathbf{u}_1), T(\mathbf{u}_2), \ldots, T(\mathbf{u}_n)$$

with respect to the basis B'. Thus, the matrix A that completes the link in Figure 8.4.2*a* is

$$A = \left[[T(\mathbf{u}_1)]_{B'} \mid [T(\mathbf{u}_2)]_{B'} \mid \cdots \mid [T(\mathbf{u}_n)]_{B'} \right] \tag{3}$$

We will call this the ***matrix for T relative to the bases B and B′*** and will denote it by the symbol $[T]_{B',B}$. Using this notation, Formula (3) can be written as

$$\boxed{[T]_{B',B} = \left[[T(\mathbf{u}_1)]_{B'} \mid [T(\mathbf{u}_2)]_{B'} \mid \cdots \mid [T(\mathbf{u}_n)]_{B'} \right]} \tag{4}$$

and from (1), this matrix has the property

$$\boxed{[T]_{B',B}[\mathbf{x}]_B = [T(\mathbf{x})]_{B'}} \tag{5}$$

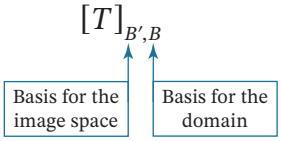


FIGURE 8.4.3

Remark Observe that in the notation $[T]_{B',B}$ the right subscript is a basis for the domain of T, and the left subscript is a basis for the image space of T (**Figure 8.4.3**). Moreover, observe how the subscript B seems to "cancel out" in Formula (5) (**Figure 8.4.4**).

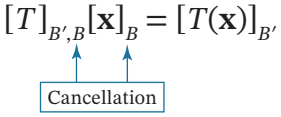


FIGURE 8.4.4

We leave it as an exercise to show that in the special case where $T_C : R^n \rightarrow R^m$ is multiplication by C, and where S and S' are the *standard bases* for R^n and R^m, respectively, then

$$[T_C]_{S',S} = C \tag{6}$$

EXAMPLE 1 | Matrix for a Linear Transformation

Let $T : P_1 \rightarrow P_2$ be the linear transformation defined by

$$T(p(x)) = xp(x)$$

Find the matrix for T with respect to the standard bases

$$B = \{\mathbf{u}_1, \mathbf{u}_2\} \quad \text{and} \quad B' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$$

That is,

$$\mathbf{u}_1 = 1, \quad \mathbf{u}_2 = x; \qquad \mathbf{v}_1 = 1, \quad \mathbf{v}_2 = x, \quad \mathbf{v}_3 = x^2$$

Solution From the given formula for T we obtain

$$T(\mathbf{u}_1) = T(1) = (x)(1) = x$$
$$T(\mathbf{u}_2) = T(x) = (x)(x) = x^2$$

By inspection, the coordinate vectors for $T(\mathbf{u}_1)$ and $T(\mathbf{u}_2)$ relative to B' are

$$[T(\mathbf{u}_1)]_{B'} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad [T(\mathbf{u}_2)]_{B'} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

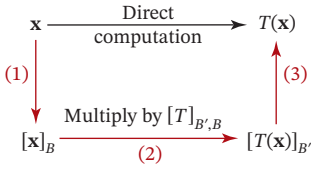
Thus, the matrix for T with respect to B and B' is

$$[T]_{B',B} = \left[[T(\mathbf{u}_1)]_{B'} \mid [T(\mathbf{u}_2)]_{B'} \right] = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

EXAMPLE 2 | The Three-Step Procedure

Let $T : P_1 \rightarrow P_2$ be the linear transformation in Example 1, and use the three-step procedure illustrated in the following figure to perform the computation

$$T(a + bx) = x(a + bx) = ax + bx^2$$



Solution

Step 1. The coordinate vector for $\mathbf{x} = a + bx$ relative to the basis $B = \{1, x\}$ is

$$[\mathbf{x}]_B = \begin{bmatrix} a \\ b \end{bmatrix}$$

Step 2. Multiplying $[\mathbf{x}]_B$ by the matrix $[T]_{B',B}$ found in Example 1 we obtain

$$[T]_{B',B}[\mathbf{x}]_B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ a \\ b \end{bmatrix} = [T(\mathbf{x})]_{B'}$$

Step 3. Reconstructing $T(\mathbf{x}) = T(a + bx)$ from $[T(\mathbf{x})]_{B'}$ we obtain

$$T(a + bx) = 0 + ax + bx^2 = ax + bx^2$$

EXAMPLE 3 | Matrix for a Linear Transformation

Let $T : R^2 \rightarrow R^3$ be the linear transformation defined by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_2 \\ -5x_1 + 13x_2 \\ -7x_1 + 16x_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -5 & 13 \\ -7 & 16 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Find the matrix for the transformation T with respect to the bases $B = \{\mathbf{u}_1, \mathbf{u}_2\}$ for R^2 and $B' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for R^3, where

$$\mathbf{u}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}; \qquad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

Solution From the formula for T,

$$T(\mathbf{u}_1) = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}, \quad T(\mathbf{u}_2) = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$$

Expressing these vectors as linear combinations of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$, we obtain (verify)

$$T(\mathbf{u}_1) = \mathbf{v}_1 - 2\mathbf{v}_3, \quad T(\mathbf{u}_2) = 3\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3$$

Thus,

$$[T(\mathbf{u}_1)]_{B'} = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \quad [T(\mathbf{u}_2)]_{B'} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}$$

so

$$[T]_{B',B} = \left[[T(\mathbf{u}_1)]_{B'} \mid [T(\mathbf{u}_2)]_{B'} \right] = \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ -2 & -1 \end{bmatrix}$$

Remark Example 3 illustrates that a fixed linear transformation generally has multiple representations, each depending on the bases chosen. In this case the matrices

$$[T]_{S',S} = \begin{bmatrix} 0 & 1 \\ -5 & 13 \\ -7 & 16 \end{bmatrix} \quad \text{and} \quad [T]_{B',B} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ -2 & -1 \end{bmatrix}$$

both represent the transformation T, the first relative to the standard bases, S and S', for R^2 and R^3, and the second relative to the bases B and B' stated in the example.

Matrices of Linear Operators

In the special case where $V = W$ (so that $T : V \to V$ is a linear operator), it is usual to take $B = B'$ when constructing a matrix for T. In this case the resulting matrix is called the ***matrix for T relative to the basis B*** and is usually denoted by $[T]_B$ rather than $[T]_{B,B}$. If $B = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$, then Formulas (4) and (5) become

$$\boxed{[T]_B = [[T(\mathbf{u}_1)]_B \mid [T(\mathbf{u}_2)]_B \mid \cdots \mid [T(\mathbf{u}_n)]_B]} \tag{7}$$

$$\boxed{[T]_B[\mathbf{x}]_B = [T(\mathbf{x})]_B} \tag{8}$$

Phrased informally, Formulas (7) and (8) state that *the matrix for T, when multiplied by the coordinate vector for $\mathbf{x}$, produces the coordinate vector for $T(\mathbf{x})$.*

We leave it for you to verify that if $T : R^n \to R^n$ is a matrix operator, say multiplication by A, and B is the standard basis for R^n, then Formula (7) simplifies to

$$[T_A]_B = A \tag{9}$$

Matrices of Identity Operators

Recall that the identity operator $I : V \to V$ maps each vector in a vector space V into itself, that is, $I(\mathbf{x}) = \mathbf{x}$ for every vector $\mathbf{x}$ in V. The following example shows that if V is n-dimensional, then the matrix for I relative to *any* basis B for V is the $n \times n$ identity matrix.

EXAMPLE 4 | Matrices of Identity Operators

If $B = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ is a basis for an n-dimensional vector space V, and if $I : V \to V$ is the identity operator on V, then

$$I(\mathbf{u}_1) = \mathbf{u}_1, \quad I(\mathbf{u}_2) = \mathbf{u}_2, \ldots, \quad I(\mathbf{u}_n) = \mathbf{u}_n$$

Therefore,

$$[I]_B = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = I$$

$[I(\mathbf{u}_1)]_B \quad [I(\mathbf{u}_2)]_B \quad [I(\mathbf{u}_n)]_B$

EXAMPLE 5 | Linear Operator on P_2

Let $T : P_2 \to P_2$ be the linear operator defined by

$$T(p(x)) = p(3x - 5)$$

that is, $T(c_0 + c_1x + c_2x^2) = c_0 + c_1(3x - 5) + c_2(3x - 5)^2$.

(*a*) Find $[T]_B$ relative to the basis $B = \{1, x, x^2\}$.

(*b*) Use the indirect procedure to compute $T(1 + 2x + 3x^2)$.

(*c*) Check the result in (b) by computing $T(1 + 2x + 3x^2)$ directly.

Solution (*a*) From the formula for T,

$$T(1) = 1, \quad T(x) = 3x - 5, \quad T(x^2) = (3x - 5)^2 = 9x^2 - 30x + 25$$

so

$$[T(1)]_B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad [T(x)]_B = \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix}, \quad [T(x^2)]_B = \begin{bmatrix} 25 \\ -30 \\ 9 \end{bmatrix}$$

Thus,

$$[T]_B = \begin{bmatrix} 1 & -5 & 25 \\ 0 & 3 & -30 \\ 0 & 0 & 9 \end{bmatrix}$$

Solution (*b*)

Step 1. The coordinate vector for $\mathbf{p} = 1 + 2x + 3x^2$ relative to the basis $B = \{1, x, x^2\}$ is

$$[\mathbf{p}]_B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Step 2. Multiplying $[\mathbf{p}]_B$ by the matrix $[T]_B$ found in part (a) we obtain

$$[T]_B[\mathbf{p}]_B = \begin{bmatrix} 1 & -5 & 25 \\ 0 & 3 & -30 \\ 0 & 0 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 66 \\ -84 \\ 27 \end{bmatrix} = [T(\mathbf{p})]_B$$

Step 3. Reconstructing $T(\mathbf{p}) = T(1 + 2x + 3x^2)$ from $[T(\mathbf{p})]_B$ we obtain

$$T(1 + 2x + 3x^2) = 66 - 84x + 27x^2$$

Solution (*c*) By direct computation,

$$\begin{aligned} T(1 + 2x + 3x^2) &= 1 + 2(3x - 5) + 3(3x - 5)^2 \\ &= 1 + 6x - 10 + 27x^2 - 90x + 75 \\ &= 66 - 84x + 27x^2 \end{aligned}$$

which agrees with the result in (b).

Matrices of Compositions and Inverse Transformations

We will conclude this section by mentioning two theorems without proof that are generalizations of earlier results.

Theorem 8.4.1

If $T_1 : U \to V$ and $T_2 : V \to W$ are linear transformations, and if B, B'', and B' are bases for U, V, and W, respectively, then

$$[T_2 \circ T_1]_{B',B} = [T_2]_{B',B''}[T_1]_{B'',B} \tag{10}$$

Theorem 8.4.2

If $T : V \to V$ is a linear operator, and if B is a basis for V, then the following are equivalent.

(a) T is one-to-one.

(b) $[T]_B$ is invertible.

Moreover, when these equivalent conditions hold,

$$[T^{-1}]_B = [T]_B^{-1} \tag{11}$$

Remark In (10), observe how the interior subscript B'' (the basis for the intermediate space V) seems to "cancel out," leaving only the bases for the domain and image space of the composition as subscripts (**Figure 8.4.5**). This "cancellation" of interior subscripts suggests the following extension of Formula (10) to compositions of three linear transformations (**Figure 8.4.6**):

$$\boxed{[T_3 \circ T_2 \circ T_1]_{B',B} = [T_3]_{B',B'''}[T_2]_{B''',B''}[T_1]_{B'',B}} \tag{12}$$

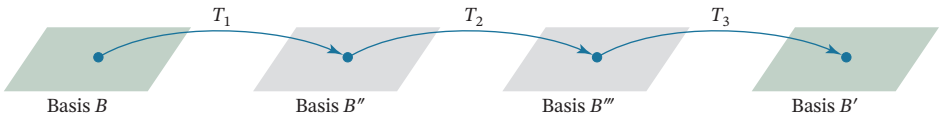


FIGURE 8.4.6

The following example illustrates Theorem 8.4.1.

$$[T_2 \circ T_1]_{B',B} = [T_2]_{B',B''}[T_1]_{B'',B}$$

Cancellation

FIGURE 8.4.5

EXAMPLE 6 | Composition

Let $T_1 : P_1 \to P_2$ be the linear transformation defined by

$$T_1(p(x)) = xp(x)$$

and let $T_2 : P_2 \to P_2$ be the linear operator defined by

$$T_2(p(x)) = p(3x - 5)$$

Then the composition $(T_2 \circ T_1) : P_1 \to P_2$ is given by

$$(T_2 \circ T_1)(p(x)) = T_2(T_1(p(x))) = T_2(xp(x)) = (3x - 5)p(3x - 5)$$

Thus, if $p(x) = c_0 + c_1x$, then

$$\begin{aligned}(T_2 \circ T_1)(c_0 + c_1x) &= (3x - 5)(c_0 + c_1(3x - 5)) \\ &= c_0(3x - 5) + c_1(3x - 5)^2 \end{aligned} \tag{13}$$

In this example, P_1 plays the role of U in Theorem 8.4.1, and P_2 plays the roles of both V and W; thus we can take $B' = B''$ in (10) so that the formula simplifies to

$$[T_2 \circ T_1]_{B',B} = [T_2]_{B'}[T_1]_{B',B} \tag{14}$$

Let us choose $B = \{1, x\}$ to be the basis for P_1 and choose $B' = \{1, x, x^2\}$ to be the basis for P_2. We showed in Examples 1 and 5 that

$$[T_1]_{B',B} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad [T_2]_{B'} = \begin{bmatrix} 1 & -5 & 25 \\ 0 & 3 & -30 \\ 0 & 0 & 9 \end{bmatrix}$$

Thus, it follows from (14) that

$$[T_2 \circ T_1]_{B',B} = \begin{bmatrix} 1 & -5 & 25 \\ 0 & 3 & -30 \\ 0 & 0 & 9 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -5 & 25 \\ 3 & -30 \\ 0 & 9 \end{bmatrix} \tag{15}$$

As a check, we will calculate $[T_2 \circ T_1]_{B',B}$ directly from Formula (4). Since $B = \{1, x\}$, it follows from Formula (4) with $\mathbf{u}_1 = 1$ and $\mathbf{u}_2 = x$ that

$$[T_2 \circ T_1]_{B',B} = \left[[(T_2 \circ T_1)(1)]_{B'} \mid [(T_2 \circ T_1)(x)]_{B'} \right] \tag{16}$$

Using (13) yields

$$(T_2 \circ T_1)(1) = 3x - 5 \quad \text{and} \quad (T_2 \circ T_1)(x) = (3x - 5)^2 = 9x^2 - 30x + 25$$

From this and the fact that $B' = \{1, x, x^2\}$, it follows that

$$[(T_2 \circ T_1)(1)]_{B'} = \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} \quad \text{and} \quad [(T_2 \circ T_1)(x)]_{B'} = \begin{bmatrix} 25 \\ -30 \\ 9 \end{bmatrix}$$

Substituting in (16) yields

$$[T_2 \circ T_1]_{B',B} = \begin{bmatrix} -5 & 25 \\ 3 & -30 \\ 0 & 9 \end{bmatrix}$$

which agrees with (15).

Exercise Set 8.4

1. Let $T : P_2 \to P_3$ be the linear transformation defined by $T(p(x)) = xp(x)$.

a. Find the matrix for T relative to the standard bases

$$B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} \quad \text{and} \quad B' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$$

where

$$\mathbf{u}_1 = 1, \quad \mathbf{u}_2 = x, \quad \mathbf{u}_3 = x^2$$
$$\mathbf{v}_1 = 1, \quad \mathbf{v}_2 = x, \quad \mathbf{v}_3 = x^2, \quad \mathbf{v}_4 = x^3$$

b. Verify that the matrix $[T]_{B',B}$ obtained in part (a) satisfies Formula (5) for every vector $\mathbf{x} = c_0 + c_1x + c_2x^2$ in P_2.

2. Let $T : P_2 \to P_1$ be the linear transformation defined by

$$T(a_0 + a_1x + a_2x^2) = (a_0 + a_1) - (2a_1 + 3a_2)x$$

a. Find the matrix for the linear transformation T relative to the standard bases $B = \{1, x, x^2\}$ and $B' = \{1, x\}$ for P_2 and P_1.

b. Verify that the matrix $[T]_{B',B}$ obtained in part (a) satisfies Formula (5) for every vector $\mathbf{x} = c_0 + c_1x + c_2x^2$ in P_2.

3. Let $T : P_2 \to P_2$ be the linear operator defined by

$$T(a_0 + a_1x + a_2x^2) = a_0 + a_1(x - 1) + a_2(x - 1)^2$$

a. Find the matrix for the linear transformation T relative to the standard basis $B = \{1, x, x^2\}$ for P_2.

b. Verify that the matrix $[T]_B$ obtained in part (a) satisfies Formula (8) for every vector $\mathbf{x} = a_0 + a_1x + a_2x^2$ in P_2.

4. Let $T : R^2 \to R^2$ be the linear operator defined by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 - x_2 \\ x_1 + x_2 \end{bmatrix}$$

and let $B = \{\mathbf{u}_1, \mathbf{u}_2\}$ be the basis for which

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{u}_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

a. Find $[T]_B$.

b. Verify that Formula (8) holds for every vector $\mathbf{x}$ in R^2.

5. Let $T : R^2 \to R^3$ be defined by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 \\ -x_1 \\ 0 \end{bmatrix}$$

a. Find the matrix $[T]_{B',B}$ relative to the bases $B = \{\mathbf{u}_1, \mathbf{u}_2\}$ and $B' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

b. Verify that Formula (5) holds for every vector in R^2.

6. Let $T : R^3 \to R^3$ be the linear operator defined by

$$T(x_1, x_2, x_3) = (x_1 - x_2, x_2 - x_1, x_1 - x_3)$$

a. Find the matrix for the linear transformation T with respect to the basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = (1, 0, 1), \quad \mathbf{v}_2 = (0, 1, 1), \quad \mathbf{v}_3 = (1, 1, 0)$$

b. Verify that Formula (8) holds for every vector in R^3.

c. Is T one-to-one? If so, find the matrix of T^{-1} with respect to the basis B.

7. Let $T : P_2 \to P_2$ be the linear operator $T(p(x)) = p(2x+1)$, that is,

$$T(c_0 + c_1x + c_2x^2) = c_0 + c_1(2x+1) + c_2(2x+1)^2$$

a. Find $[T]_B$ with respect to the basis $B = \{1, x, x^2\}$.

b. Use the three-step procedure illustrated in Example 2 to compute $T(2 - 3x + 4x^2)$.

c. Check the result obtained in part (b) by computing $T(2 - 3x + 4x^2)$ directly.

8. Let $T : P_2 \to P_3$ be the linear transformation defined by $T(p(x)) = xp(x-3)$, that is,

$$T(c_0 + c_1x + c_2x^2) = x(c_0 + c_1(x-3) + c_2(x-3)^2)$$

a. Find $[T]_{B',B}$ relative to the bases $B = \{1, x, x^2\}$ and $B' = \{1, x, x^2, x^3\}$.

b. Use the three-step procedure illustrated in Example 2 to compute $T(1 + x - x^2)$.

c. Check the result obtained in part (b) by computing $T(1 + x - x^2)$ directly.

9. Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$, and let $A = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$ be the matrix for $T : R^2 \to R^2$ relative to the basis $B = \{\mathbf{v}_1, \mathbf{v}_2\}$.

a. Find $[T(\mathbf{v}_1)]_B$ and $[T(\mathbf{v}_2)]_B$.

b. Find $T(\mathbf{v}_1)$ and $T(\mathbf{v}_2)$.

c. Find a formula for $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right)$.

d. Use the formula obtained in (c) to compute $T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$.

10. Let $A = \begin{bmatrix} 3 & -2 & 1 & 0 \\ 1 & 6 & 2 & 1 \\ -3 & 0 & 7 & 1 \end{bmatrix}$ be the matrix for $T : R^4 \to R^3$ relative to the bases $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ and $B' = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 4 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 6 \\ 9 \\ 4 \\ 2 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 0 \\ 8 \\ 8 \end{bmatrix}, \quad \mathbf{w}_2 = \begin{bmatrix} -7 \\ 8 \\ 1 \end{bmatrix}, \quad \mathbf{w}_3 = \begin{bmatrix} -6 \\ 9 \\ 1 \end{bmatrix}$$

a. Find $[T(\mathbf{v}_1)]_{B'}$, $[T(\mathbf{v}_2)]_{B'}$, $[T(\mathbf{v}_3)]_{B'}$, and $[T(\mathbf{v}_4)]_{B'}$.

b. Find $T(\mathbf{v}_1)$, $T(\mathbf{v}_2)$, $T(\mathbf{v}_3)$, and $T(\mathbf{v}_4)$.

c. Find a formula for $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}\right)$.

d. Use the formula obtained in (c) to compute $T\left(\begin{bmatrix} 2 \\ 2 \\ 0 \\ 0 \end{bmatrix}\right)$.

11. Let $A = \begin{bmatrix} 1 & 3 & -1 \\ 2 & 0 & 5 \\ 6 & -2 & 4 \end{bmatrix}$ be the matrix for $T : P_2 \to P_2$ with respect to the basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where $\mathbf{v}_1 = 3x + 3x^2$, $\mathbf{v}_2 = -1 + 3x + 2x^2$, $\mathbf{v}_3 = 3 + 7x + 2x^2$.

a. Find $[T(\mathbf{v}_1)]_B$, $[T(\mathbf{v}_2)]_B$, and $[T(\mathbf{v}_3)]_B$.

b. Find $T(\mathbf{v}_1)$, $T(\mathbf{v}_2)$, and $T(\mathbf{v}_3)$.

c. Find a formula for $T(a_0 + a_1x + a_2x^2)$.

d. Use the formula obtained in (c) to compute $T(1 + x^2)$.

12. Let $T_1 : P_1 \to P_2$ be the linear transformation defined by

$$T_1(p(x)) = xp(x)$$

and let $T_2 : P_2 \to P_2$ be the linear operator defined by

$$T_2(p(x)) = p(2x+1)$$

Let $B = \{1, x\}$ and $B' = \{1, x, x^2\}$ be the standard bases for P_1 and P_2.

a. Find $[T_2 \circ T_1]_{B',B}$, $[T_2]_{B'}$, and $[T_1]_{B',B}$.

b. State a formula relating the matrices in part (a).

c. Verify that the matrices in part (a) satisfy the formula you stated in part (b).

13. Let $T_1 : P_1 \to P_2$ be the linear transformation defined by

$$T_1(c_0 + c_1x) = 2c_0 - 3c_1x$$

and let $T_2 : P_2 \to P_3$ be the linear transformation defined by

$$T_2(c_0 + c_1x + c_2x^2) = 3c_0x + 3c_1x^2 + 3c_2x^3$$

Let $B = \{1, x\}$, $B'' = \{1, x, x^2\}$, and $B' = \{1, x, x^2, x^3\}$.

a. Find $[T_2 \circ T_1]_{B',B}$, $[T_2]_{B',B''}$, and $[T_1]_{B'',B}$.

b. State a formula relating the matrices in part (a).

c. Verify that the matrices in part (a) satisfy the formula you stated in part (b).

14. Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ be a basis for a vector space V. Find the matrix with respect to B for the linear operator $T : V \to V$ defined by $T(\mathbf{v}_1) = \mathbf{v}_2$, $T(\mathbf{v}_2) = \mathbf{v}_3$, $T(\mathbf{v}_3) = \mathbf{v}_4$, $T(\mathbf{v}_4) = \mathbf{v}_1$.

15. Let $T : P_2 \to M_{22}$ be the linear transformation defined by

$$T(\mathbf{p}) = \begin{bmatrix} p(0) & p(1) \\ p(-1) & p(0) \end{bmatrix}$$

let B be the standard basis for M_{22}, and let $B' = \{1, x, x^2\}$, $B'' = \{1, 1+x, 1+x^2\}$ be bases for P_2.

a. Find $[T]_{B,B'}$ and $[T]_{B,B''}$.

b. For the matrices obtained in part (a), find

$$T(2 + 2x + x^2)$$

using the three-step procedure illustrated in Example 2.

c. Check the results obtained in part (b) by computing $T(2 + 2x + x^2)$ directly.

16. Let $T : M_{22} \rightarrow R^2$ be the linear transformation given by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a+b+c \\ d \end{bmatrix}$$

and let B be the standard basis for M_{22}, B' the standard basis for R^2, and

$$B'' = \left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \end{bmatrix}\right\}$$

a. Find $[T]_{B',B}$ and $[T]_{B'',B}$.

b. Compute $T\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\right)$ using the three-step procedure that was illustrated in Example 2 for both matrices found in part (a).

c. Check the results obtained in part (b) by computing $T\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\right)$ directly.

17. **(*Calculus required*)** Let $D : P_2 \rightarrow P_2$ be the differentiation operator $D(\mathbf{p}) = p'(x)$.

a. Find the matrix for D relative to the basis $B = \{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ for P_2 in which $\mathbf{p}_1 = 1,\ \mathbf{p}_2 = x,\ \mathbf{p}_3 = x^2$.

b. Use the matrix in part (a) to compute $D(6 - 6x + 24x^2)$.

18. **(*Calculus required*)** Let $D : P_2 \rightarrow P_2$ be the differentiation operator $D(\mathbf{p}) = p'(x)$.

a. Find the matrix for D relative to the basis $B = \{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ for P_2 in which $\mathbf{p}_1 = 2,\ \mathbf{p}_2 = 2 - 3x,\ \mathbf{p}_3 = 2 - 3x + 8x^2$.

b. Use the matrix in part (a) to compute $D(6 - 6x + 24x^2)$.

19. **(*Calculus required*)** Let V be the vector space of real-valued functions defined on the interval $(-\infty, \infty)$, and let $D : V \rightarrow V$ be the differentiation operator.

a. Find the matrix for D relative to the basis $B = \{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3\}$ for V in which $\mathbf{f}_1 = 1,\ \mathbf{f}_2 = \sin x,\ \mathbf{f}_3 = \cos x$.

b. Use the matrix in part (a) to compute

$$D(2 + 3\sin x - 4\cos x)$$

20. Let V be a four-dimensional vector space with basis B, let W be a seven-dimensional vector space with basis B', and let $T : V \rightarrow W$ be a linear transformation. Identify the four vector spaces that contain the vectors at the corners of the accompanying diagram.

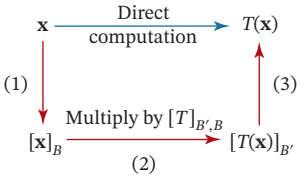


FIGURE Ex-20

21. In each part, fill in the missing part of the equation.

a. $[T_2 \circ T_1]_{B',B} = [T_2]__?__[T_1]_{B'',B}$

b. $[T_3 \circ T_2 \circ T_1]_{B',B} = [T_3]__?__[T_2]_{B''',B''}[T_1]_{B'',B}$

Working with Proofs

22. Prove: If $T : V \rightarrow W$ is the zero transformation, then the matrix for T with respect to any bases for V and W is a zero matrix.

23. Prove: If B and B' are the standard bases for R^n and R^m, respectively, then the matrix for a linear transformation $T : R^n \rightarrow R^m$ relative to the bases B and B' is the standard matrix for T.

True-False Exercises

TF. In parts **(a)–(e)** determine whether the statement is true or false, and justify your answer.

a. If the matrix of a linear transformation $T : V \rightarrow W$ relative to some bases of V and W is $\begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix}$, then there is a nonzero vector $\mathbf{x}$ in V such that $T(\mathbf{x}) = 2\mathbf{x}$.

b. If the matrix of a linear transformation $T : V \rightarrow W$ relative to bases for V and W is $\begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix}$, then there is a nonzero vector $\mathbf{x}$ in V such that $T(\mathbf{x}) = 4\mathbf{x}$.

c. If the matrix of a linear transformation $T : V \rightarrow W$ relative to certain bases for V and W is $\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$, then T is one-to-one.

d. If $S : V \rightarrow V$ and $T : V \rightarrow V$ are linear operators and B is a basis for V, then the matrix of $S \circ T$ relative to B is $[T]_B[S]_B$.

e. If $T : V \rightarrow V$ is an invertible linear operator and B is a basis for V, then the matrix for T^{-1} relative to B is $[T]_B^{-1}$.

8.5 Similarity

The matrix for a linear operator $T: V \to V$ depends on the basis selected for V. One of the fundamental problems of linear algebra is to choose a basis for V that makes the matrix for T as simple as possible—a diagonal or a triangular matrix, for example. In this section we will study this problem.

Simple Matrices for Linear Operators

Standard bases do not necessarily produce the simplest matrices for linear operators. For example, consider the matrix operator $T: R^2 \to R^2$ whose matrix relative to the standard basis $B = \{\mathbf{e}_1, \mathbf{e}_2\}$ for R^2 is

$$[T]_B = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \tag{1}$$

Let us compare this matrix to the matrix $[T]_{B'}$ for the same operator T but relative to the basis $B' = \{\mathbf{u}_1', \mathbf{u}_2'\}$ for R^2 in which

$$\mathbf{u}_1' = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2' = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \tag{2}$$

Since

$$T(\mathbf{u}_1') = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 2\mathbf{u}_1' \quad \text{and} \quad T(\mathbf{u}_2') = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} = 3\mathbf{u}_2'$$

it follows that

$$[T(\mathbf{u}_1')]_{B'} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad \text{and} \quad [T(\mathbf{u}_2')]_{B'} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

so the matrix for T relative to the basis B' is

$$[T]_{B'} = \left[T(\mathbf{u}_1')_{B'} \mid T(\mathbf{u}_2')_{B'}\right] = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

This matrix, being diagonal, has a simpler form than $[T]_B$ and conveys clearly that the operator T scales $\mathbf{u}_1'$ by a factor of 2 and $\mathbf{u}_2'$ by a factor of 3, information that is not immediately evident from $[T]_B$.

One of the major themes in more advanced linear algebra courses is to determine the "simplest possible form" that can be obtained for the matrix of a linear operator by choosing the basis appropriately. Sometimes it is possible to obtain a diagonal matrix (as above, for example), whereas other times one must settle for a triangular matrix or some other form. We will only be able to touch on this important topic in this text.

The problem of finding a basis that produces the simplest possible matrix for a linear operator $T: V \to V$ can be attacked by first finding a matrix for T relative to *any* basis, typically a standard basis, where applicable, and then changing the basis in a way that simplifies the matrix. Before pursuing this idea, it will be helpful to revisit some concepts about changing bases.

A New View of Transition Matrices

Recall from Formulas (9) and (10) of Section 4.7 that if $B = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ and $B' = \{\mathbf{u}_1', \mathbf{u}_2', \ldots, \mathbf{u}_n'\}$ are bases for a vector space V, then the transition matrices from B to B' and from B' to B are

$$P_{B \to B'} = \left[[\mathbf{u}_1]_{B'} \mid [\mathbf{u}_2]_{B'} \mid \cdots \mid [\mathbf{u}_n]_{B'}\right] \tag{3}$$

$$P_{B' \to B} = \left[[\mathbf{u}_1']_B \mid [\mathbf{u}_2']_B \mid \cdots \mid [\mathbf{u}_n']_B\right] \tag{4}$$

where the matrices $P_{B \to B'}$ and $P_{B' \to B}$ are inverses of each other. We also showed in Formulas (11) and (12) of that section that if $\mathbf{v}$ is any vector in V, then

$$P_{B \to B'}[\mathbf{v}]_B = [\mathbf{v}]_{B'} \tag{5}$$

$$P_{B' \to B}[\mathbf{v}]_{B'} = [\mathbf{v}]_B \tag{6}$$

The following theorem shows that transition matrices in Formulas (3) and (4) can be viewed as matrices for identity operators.

Theorem 8.5.1

If B and B' are bases for a finite-dimensional vector space V, and if $I : V \to V$ is the identity operator on V, then

$$P_{B \to B'} = [I]_{B',B} \quad \text{and} \quad P_{B' \to B} = [I]_{B,B'}$$

Proof Suppose that $B = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ and $B' = \{\mathbf{u}'_1, \mathbf{u}'_2, \ldots, \mathbf{u}'_n\}$ are bases for V. Using the fact that $I(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v}$ in V, it follows from Formula (4) of Section 8.4 that

$$\begin{aligned} [I]_{B',B} &= \left[[I(\mathbf{u}_1)]_{B'} \mid [I(\mathbf{u}_2)]_{B'} \mid \cdots \mid [I(\mathbf{u}_n)]_{B'} \right] \\ &= \left[[\mathbf{u}_1]_{B'} \mid [\mathbf{u}_2]_{B'} \mid \cdots \mid [\mathbf{u}_n]_{B'} \right] \\ &= P_{B \to B'} \quad \textbf{[Formula (3)]} \end{aligned}$$

The proof that $[I]_{B,B'} = P_{B' \to B}$ is similar. ■

Effect of Changing Bases on Matrices of Linear Operators

We are now ready to consider the main problem in this section.

Problem

If B and B' are two bases for a finite-dimensional vector space V, and if $T : V \to V$ is a linear operator, what relationship, if any, exists between the matrices $[T]_B$ and $[T]_{B'}$?

The answer to this question can be obtained by considering the composition of the three linear operators on V pictured in **Figure 8.5.1**.

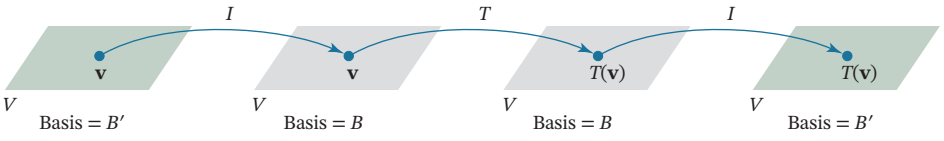


FIGURE 8.5.1

In this figure, $\mathbf{v}$ is first mapped into itself by the identity operator, then $\mathbf{v}$ is mapped into $T(\mathbf{v})$ by T, and then $T(\mathbf{v})$ is mapped into itself by the identity operator. All four vector spaces involved in the composition are the same (namely, V), but the bases for the spaces vary. Since the starting vector is $\mathbf{v}$ and the final vector is $T(\mathbf{v})$, the composition produces the same result as applying T directly; that is,

$$T = I \circ T \circ I \tag{7}$$

If, as illustrated in **Figure 8.5.1**, the first and last vector spaces are assigned the basis B' and the middle two spaces are assigned the basis B, then it follows from (7) and Formula (12) of Section 8.4 (with an appropriate adjustment to the names of the bases) that

$$[T]_{B',B'} = [I \circ T \circ I]_{B',B'} = [I]_{B',B}[T]_{B,B}[I]_{B,B'} \tag{8}$$

or, in simpler notation,

$$[T]_{B'} = [I]_{B',B}[T]_B[I]_{B,B'} \tag{9}$$

We can simplify this formula even further by using Theorem 8.5.1 to rewrite it as

$$\boxed{[T]_{B'} = P_{B\to B'}[T]_B P_{B'\to B}} \tag{10}$$

In summary, we have the following theorem.

Theorem 8.5.2

Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V, and let B and B' be bases for V. Then

$$[T]_{B'} = P^{-1}[T]_B P \tag{11}$$

where $P = P_{B'\to B}$ and $P^{-1} = P_{B\to B'}$.

Warning When applying Theorem 8.5.2, it is easy to forget whether $P = P_{B'\to B}$ (correct) or $P = P_{B\to B'}$ (incorrect). It may help to use the diagram in **Figure 8.5.2** and observe that the *exterior* subscripts of the transition matrices match the subscript of the matrix they enclose.

$$[T]_{B'} = P_{B\to B'}\,[T]_B\,P_{B'\to B}$$

Exterior subscripts

FIGURE 8.5.2

In the terminology of Definition 1 of Section 5.2, Theorem 8.5.2 tells us that matrices representing the same linear operator relative to different bases must be similar. The following theorem, which we state without proof, shows that the converse of Theorem 8.5.2 is also true.

Theorem 8.5.3

If V is a finite-dimensional vector space, then two matrices A and B represent the same linear operator (but possibly with respect to different bases) if and only if they are similar. Moreover, if $B = P^{-1}AP$, then P is the transition matrix from the basis used for B to the basis used for A.

EXAMPLE 1 | Similar Matrices Represent the Same Linear Operator

We showed at the beginning of this section that the matrices

$$C = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

represent the same linear operator $T : R^2 \to R^2$ with respect to the appropriate bases. Verify that these matrices are similar by finding a matrix P for which $D = P^{-1}CP$.

Solution We need to find the transition matrix

$$P = P_{B'\to B} = [[\mathbf{u}_1']_B \mid [\mathbf{u}_2']_B]$$

where $B' = \{\mathbf{u}_1', \mathbf{u}_2'\}$ is the basis for R^2 given by (2) and $B = \{\mathbf{e}_1, \mathbf{e}_2\}$ is the standard basis for R^2. We see by inspection that

$$\mathbf{u}_1' = \mathbf{e}_1 + \mathbf{e}_2$$
$$\mathbf{u}_2' = \mathbf{e}_1 + 2\mathbf{e}_2$$

from which it follows that

$$[\mathbf{u}_1']_B = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad [\mathbf{u}_2']_B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Thus,

$$P = P_{B' \to B} = [[\mathbf{u}_1']_B \mid [\mathbf{u}_2']_B] = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

We leave it for you to verify that

$$P^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

and hence that

$$\underset{D}{\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}} = \underset{P^{-1}}{\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}} \underset{C}{\begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}} \underset{P}{\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}}$$

Similarity Invariants

Recall from Section 5.2 that a property of a square matrix is called a *similarity invariant* if that property is shared by all similar matrices. In **Table 1** of that section we listed the most important similarity invariants. Since we know from Theorem 8.5.3 that two matrices are similar if and only if they represent the same linear operator $T : V \to V$, it follows that if B and B' are bases for V, then every similarity invariant property of $[T]_B$ is also a similarity invariant property of $[T]_{B'}$. For example, for any two bases B and B' we must have

$$\det[T]_B = \det[T]_{B'}$$

It follows from this equation that the value of the determinant depends on T, but not on the particular basis that is used to represent T in matrix form. Thus, the determinant can be regarded as a property of the linear operator T, and we can *define* the ***determinant of the linear operator T*** to be

$$\det(T) = \det[T]_B \tag{12}$$

where B is *any* basis for V. **Table 1** lists the basic similarity invariants of a linear operator $T : V \to V$.

TABLE 1 Similarity Invariants

Property	Similarity
Determinant	$[T]_B$ and $P^{-1}[T]_B P$ have the same determinant.
Invertibility	$[T]_B$ is invertible if and only if $P^{-1}[T]_B P$ is invertible.
Rank	$[T]_B$ and $P^{-1}[T]_B P$ have the same rank.
Nullity	$[T]_B$ and $P^{-1}[T]_B P$ have the same nullity.
Trace	$[T]_B$ and $P^{-1}[T]_B P$ have the same trace.
Characteristic polynomial	$[T]_B$ and $P^{-1}[T]_B P$ have the same characteristic polynomial.
Eigenvalues	$[T]_B$ and $P^{-1}[T]_B P$ have the same eigenvalues.
Eigenspace dimension	If λ is an eigenvalue of $[T]_B$ and $P^{-1}[T]_B P$, then the eigenspace of $[T]_B$ corresponding to λ and the eigenspace of $P^{-1}[T]_B P$ corresponding to λ have the same dimension.

EXAMPLE 2 | Determinant of a Linear Operator

At the beginning of this section we showed that the matrices

$$[T] = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \quad \text{and} \quad [T]_{B'} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

represent the same linear operator relative to different bases, the first relative to the standard basis $B = \{\mathbf{e}_1, \mathbf{e}_2\}$ for R^2 and the second relative to the basis $B' = \{\mathbf{u}_1', \mathbf{u}_2'\}$ for which

$$\mathbf{u}_1' = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2' = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

This means that $[T]$ and $[T]_{B'}$ must be similar matrices and hence must have the same similarity invariant properties. In particular, they must have the same determinant. The following computations confirm this.

$$\det[T] = \begin{vmatrix} 1 & 1 \\ -2 & 4 \end{vmatrix} = 6 \quad \text{and} \quad \det[T]_{B'} = \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} = 6$$

EXAMPLE 3 | Eigenvalues of a Linear Operator

Find the eigenvalues of the linear operator $T : P_2 \rightarrow P_2$ defined by

$$T(a + bx + cx^2) = -2c + (a + 2b + c)x + (a + 3c)x^2$$

Solution Because eigenvalues are similarity invariants, we can find the eigenvalues of T by choosing any basis B for P_2 and computing the eigenvalues of the matrix $[T]_B$. We leave it for you to show that the matrix for T relative to the standard basis $B = \{1, x, x^2\}$ is

$$[T]_B = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

Thus, the eigenvalues of T are $\lambda = 1$ and $\lambda = 2$ (see Example 7 of Section 5.1).

Exercise Set **8.5**

In Exercises 1–2, use a property from Table 1 to show that the matrices A and B are not similar.

1. a. $A = \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$

b. $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

2. a. $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

b. $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

3. Let $T : R^2 \rightarrow R^2$ be a linear operator, and let B and B' be bases for R^2 for which

$$[T]_B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad P_{B \rightarrow B'} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

Find the matrix for T relative to the basis B'.

4. Let $T : R^2 \rightarrow R^2$ be a linear operator, and let B and B' be bases for R^2 for which

$$[T]_B = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} \quad \text{and} \quad P_{B' \rightarrow B} = \begin{bmatrix} 4 & 5 \\ 1 & -1 \end{bmatrix}$$

Find the matrix for T relative to the basis B'.

5. Let $T : R^2 \to R^2$ be a linear operator, and let B and B' be bases for R^2 for which

$$[T]_{B'} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad P_{B \to B'} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

Find the matrix for T relative to the basis B.

6. Let $T : R^2 \to R^2$ be a linear operator, and let B and B' be bases for R^2 for which

$$[T]_{B'} = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} \quad \text{and} \quad P_{B' \to B} = \begin{bmatrix} 4 & 5 \\ 1 & -1 \end{bmatrix}$$

Find the matrix for T relative to the basis B.

In Exercises 7–14, find the matrix for T relative to the basis B, and use Theorem 8.5.2 to compute the matrix for T relative to the basis B'.

7. $T : R^2 \to R^2$ is defined by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 - 2x_2 \\ -x_2 \end{bmatrix}$$

and $B = \{\mathbf{u}_1, \mathbf{u}_2\}$ and $B' = \{\mathbf{v}_1, \mathbf{v}_2\}$, where

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}; \qquad \mathbf{v}_1 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 7 \\ 2 \end{bmatrix}$$

8. $T : R^2 \to R^2$ is defined by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 7x_2 \\ 3x_1 - 4x_2 \end{bmatrix}$$

and $B = \{\mathbf{u}_1, \mathbf{u}_2\}$ and $B' = \{\mathbf{v}_1, \mathbf{v}_2\}$, where

$$\mathbf{u}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 4 \\ -1 \end{bmatrix}; \qquad \mathbf{v}_1 = \begin{bmatrix} 18 \\ 8 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$$

9. $T : R^3 \to R^3$ is defined by

$$T(x_1, x_2, x_3) = (-2x_1 - x_2, x_1 + x_3, x_2)$$

B is the standard basis, and $B' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = (-2, 1, 0), \quad \mathbf{v}_2 = (-1, 0, 1), \quad \mathbf{v}_3 = (0, 1, 0)$$

10. $T : R^3 \to R^3$ is defined by

$$T(x_1, x_2, x_3) = (x_1 + 2x_2 - x_3, -x_2, x_1 + 7x_3)$$

B is the standard basis, and $B' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = (1, 0, 0), \quad \mathbf{v}_2 = (1, 1, 0), \quad \mathbf{v}_3 = (1, 1, 1)$$

11. $T : R^2 \to R^2$ is the rotation about the origin through an angle of 45°, B is the standard basis, and $B' = \{\mathbf{v}_1, \mathbf{v}_2\}$, where

$$\mathbf{v}_1 = \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}\right), \quad \mathbf{v}_2 = \left(-\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}\right)$$

12. $T : R^2 \to R^2$ is the shear in the x-direction by a positive factor k, B is the standard basis, and $B' = \{\mathbf{v}_1, \mathbf{v}_2\}$, where

$$\mathbf{v}_1 = (k, 1), \quad \mathbf{v}_2 = (1, 0)$$

13. $T : P_1 \to P_1$ is defined by

$$T(a_0 + a_1 x) = -a_0 + (a_0 + a_1)x$$

B is the standard basis for P_1, and $B' = \{\mathbf{q}_1, \mathbf{q}_2\}$, where

$$\mathbf{q}_1 = x + 1, \quad \mathbf{q}_2 = x - 1$$

14. $T : P_1 \to P_1$ is defined by $T(a_0 + a_1 x) = a_0 + a_1(x + 1)$, and $B = \{\mathbf{p}_1, \mathbf{p}_2\}$ and $B' = \{\mathbf{q}_1, \mathbf{q}_2\}$, where

$$\mathbf{p}_1 = 6 + 3x, \quad \mathbf{p}_2 = 10 + 2x; \qquad \mathbf{q}_1 = 2, \quad \mathbf{q}_2 = 3 + 2x$$

15. Let $T : P_2 \to P_2$ be defined by

$$T(a_0 + a_1 x + a_2 x^2) = (5a_0 + 6a_1 + 2a_2) - (a_1 + 8a_2)x + (a_0 - 2a_2)x^2$$

a. Find the eigenvalues of T.

b. Find bases for the eigenspaces of T.

16. Let $T : M_{22} \to M_{22}$ be defined by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} 2c & a + c \\ b - 2c & d \end{bmatrix}$$

a. Find the eigenvalues of T.

b. Find bases for the eigenspaces of T.

17. Since the standard basis for R^n is so simple, why would one want to represent a linear operator on R^n in another basis?

18. Find two nonzero 2×2 matrices (different from those in Exercise 1) that are not similar, and explain why they are not.

In Exercises 19–21, find the determinant and the eigenvalues of the linear operator T.

19. $T : R^2 \to R^2$, where $T(x_1, x_2) = (3x_1 - 4x_2, -x_1 + 7x_2)$

20. $T : R^3 \to R^3$, where $T(x_1, x_2, x_3) = (x_1 - x_2, x_2 - x_3, x_3 - x_1)$

21. $T : P_2 \to P_2$, where $T(p(x)) = p(x - 1)$

22. Let $T : P_4 \to P_4$ be the linear operator given by the formula $T(p(x)) = p(2x + 1)$.

a. Find a matrix for T relative to some convenient basis, and then use it to find the rank and nullity of T.

b. Use the result in part (a) to determine whether T is one-to-one.

Working with Proofs

23. Complete the proof below by justifying each step.

Hypothesis: A and B are similar matrices.

Conclusion: A and B have the same characteristic polynomial.

Proof: (1) $\det(\lambda I - B) = \det(\lambda I - P^{-1}AP)$

(2) $= \det(\lambda P^{-1}P - P^{-1}AP)$

(3) $= \det(P^{-1}(\lambda I - A)P)$

(4) $= \det(P^{-1})\det(\lambda I - A)\det(P)$

(5) $= \det(P^{-1})\det(P)\det(\lambda I - A)$

(6) $= \det(\lambda I - A)$

24. If A and B are similar matrices, say $B = P^{-1}AP$, then it follows from Exercise 23 that A and B have the same eigenvalues. Suppose that λ is one of the common eigenvalues and $\mathbf{x}$ is a corresponding eigenvector of A. See if you can find an eigenvector of B corresponding to λ (expressed in terms of λ, $\mathbf{x}$, and P).

In Exercises 25–28, prove that the stated property is a similarity invariant.

25. Trace

26. Rank

27. Nullity

28. Invertibility

29. Let λ be an eigenvalue of a linear operator $T : V \to V$. Prove that the eigenvectors of T corresponding to λ are the nonzero vectors in the kernel of $\lambda I - T$.

30. **a.** Prove that if A and B are similar matrices, then A^2 and B^2 are also similar.

b. If A^2 and B^2 are similar, must A and B be similar? Explain.

31. Let C and D be $m \times n$ matrices, and let $B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a basis for a vector space V. Prove that if $C[\mathbf{x}]_B = D[\mathbf{x}]_B$ for all $\mathbf{x}$ in V, then $C = D$.

True-False Exercises

TF. In parts **(a)–(h)** determine whether the statement is true or false, and justify your answer.

a. A matrix cannot be similar to itself.

b. If A is similar to B, and B is similar to C, then A is similar to C.

c. If A and B are similar and B is singular, then A is singular.

d. If A and B are invertible and similar, then A^{-1} and B^{-1} are similar.

e. If $T_1 : R^n \to R^n$ and $T_2 : R^n \to R^n$ are linear operators, and if $[T_1]_{B',B} = [T_2]_{B',B}$ with respect to two bases B and B' for R^n, then $T_1(\mathbf{x}) = T_2(\mathbf{x})$ for every vector $\mathbf{x}$ in R^n.

f. If $T_1 : R^n \to R^n$ is a linear operator, and if $[T_1]_B = [T_1]_{B'}$ with respect to two bases B and B' for R^n, then $B = B'$.

g. If $T : R^n \to R^n$ is a linear operator, and if $[T]_B = I_n$ with respect to some basis B for R^n, then T is the identity operator on R^n.

h. If $T : R^n \to R^n$ is a linear operator, and if $[T]_{B',B} = I_n$ with respect to two bases B and B' for R^n, then T is the identity operator on R^n.

Working with Technology

T1. Use the matrices A and P given below to construct a matrix $B = P^{-1}AP$ that is similar to A, and confirm, in accordance with Table 1, that A and B have the same determinant, trace, rank, characteristic equation, and eigenvalues.

$$A = \begin{bmatrix} -13 & -60 & -60 \\ 10 & 42 & 40 \\ -5 & -20 & -18 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & -1 \\ -1 & -1 & 0 \end{bmatrix}$$

T2. Let $T : R^3 \to R^3$ be the linear transformation whose standard matrix is the matrix A in Exercise T1. Find a basis S for R^3 for which $[T]_S$ is diagonal.

8.6 Geometry of Matrix Operators

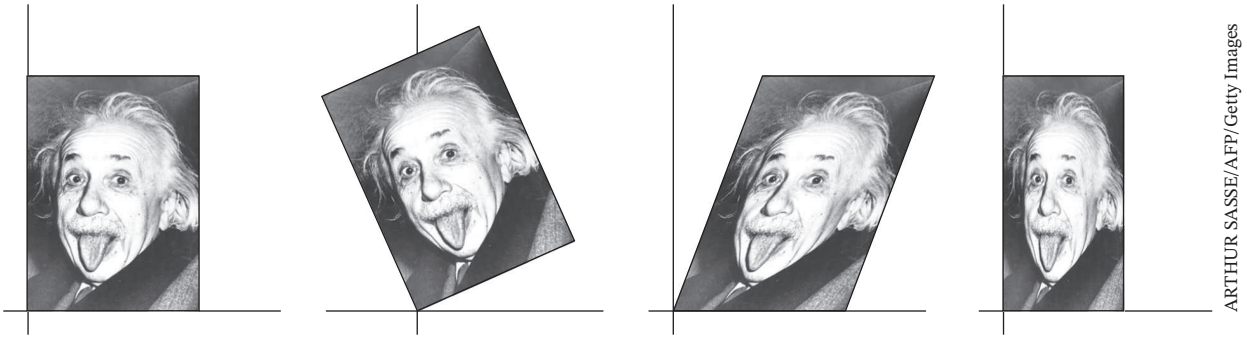
In applications such as computer graphics it is important to understand not only how linear operators on R^2 and R^3 affect individual vectors but also how they transform entire regions in two and three dimensions. In R^2, for example, one can get a sense of the effect of a linear operator on all regions in the plane by examining its effect on a unit square in the first quadrant. This will be our primary focus in this section. We will also continue our study of rotations by extending our work in Section 1.8 to three dimensions.

Computerized Transformations

Figure 8.6.1 shows a famous picture of Albert Einstein that has been transformed in various ways using matrix operators on R^2. The original image was scanned and then digitized to decompose it into a rectangular array of pixels. Those pixels were then transformed as follows:

- The program MATLAB was used to assign coordinates and a gray level to each pixel.
- The coordinates of the pixels were transformed by matrix multiplication.
- The pixels were then assigned their original gray levels to produce the transformed picture.

In computer games a perception of motion is created by using matrices to rapidly and repeatedly transform the arrays of pixels that form the visual images.



ARTHUR SASSE/AFP/Getty Images

Digitized scan

FIGURE 8.6.1

Images of Lines Under Matrix Operators

The effect of a matrix operator on R^2 can often be deduced by studying how it transforms the points that form the unit square. The following theorem, which we state without proof, shows that if an operator is invertible, then it maps each line segment in the unit square into the line segment connecting the images of its endpoints. In particular, the edges of the unit square get mapped into edges of the image (see **Figure 8.6.2** in which the edges of the unit square and the corresponding edges of its image have been numbered).

Theorem 8.6.1

If $T : R^2 \rightarrow R^2$ is multiplication by an invertible matrix, then:

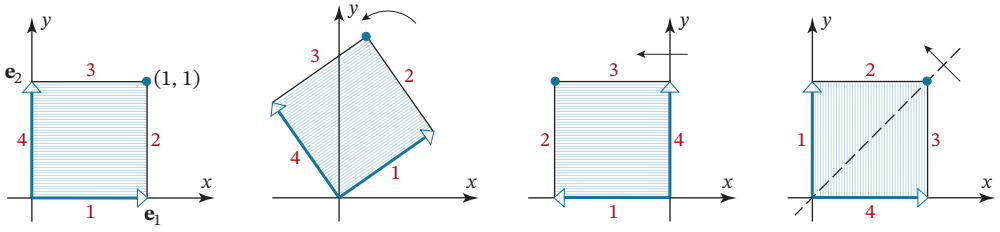
(*a*) The image of a straight line is a straight line.

(*b*) The image of a line through the origin is a line through the origin.

(*c*) The images of parallel lines are parallel lines.

(*d*) The image of the line segment joining points P and Q is the line segment joining the images of P and Q.

(*e*) The images of three points lie on a line if and only if the points themselves lie on a line.



Unit square | Unit square rotated | Unit square reflected about the y-axis | Unit square reflected about the line $y = x$

FIGURE 8.6.2

EXAMPLE 1 | Image of a Line

According to Theorem 8.6.1, the invertible matrix

$$A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

maps the line $y = 2x + 1$ into another line. Find its equation.

Solution Let (x, y) be a point on the line $y = 2x + 1$, and let (x', y') be its image under multiplication by A. Then

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

so

$$\begin{aligned} x &= x' - y' \\ y &= -2x' + 3y' \end{aligned}$$

Substituting these expressions in $y = 2x + 1$ yields

$$-2x' + 3y' = 2(x' - y') + 1$$

or, equivalently,

$$y' = \tfrac{4}{5}x' + \tfrac{1}{5}$$

EXAMPLE 2 | Transformation of the Unit Square

Sketch the image of the unit square under multiplication by the invertible matrix

$$A = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$$

Label the vertices of the image with their coordinates, and number the edges of the unit square and their corresponding images (as in Figure 8.6.2).

Solution Since

$$\begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix},$$

$$\begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

the image of the unit square is a parallelogram with vertices $(0, 0)$, $(0, 2)$, $(1, 1)$, and $(1, 3)$ (**Figure 8.6.3**).

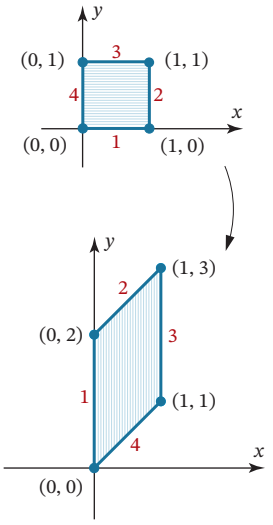
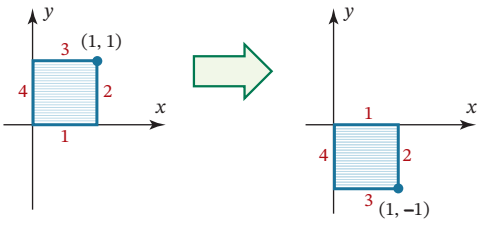
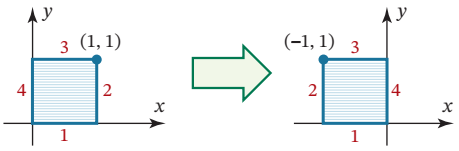
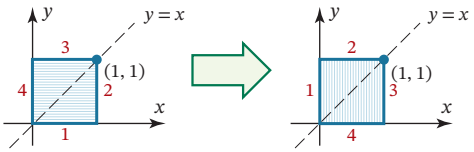


FIGURE 8.6.3

Reflections, Rotations, and Projections

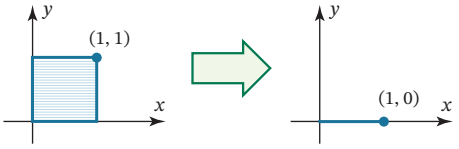
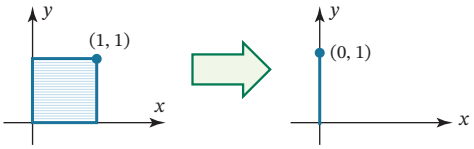
In Table 1 of Section 1.8 we obtained the standard matrices for the reflections about the x-axis, the y-axis, and the line $y = x$ in R^2. **Table 1** illustrates the effect of those transformations on the unit square.

TABLE 1

Operator	Standard Matrix	Effect on the Unit Square
Reflection about the x-axis	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	
Reflection about the y-axis	$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$	
Reflection about the line $y = x$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	

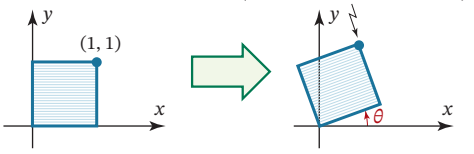
In Table 3 of Section 1.8 we obtained the standard matrices for the orthogonal projections onto the x-axis and the y-axis. **Table 2** illustrates how those projections totally flatten the unit square.

TABLE 2

Operator	Standard Matrix	Effect on the Unit Square
Orthogonal projection onto the x-axis	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	
Orthogonal projection onto the y-axis	$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$	

In Table 5 of Section 1.8 we obtained the standard matrix for a rotation about the origin through a positive angle θ. **Table 3** illustrates how such a rotation transforms the unit square.

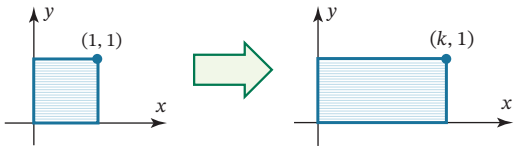
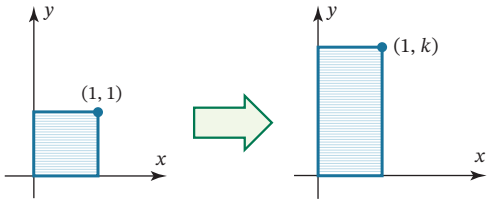
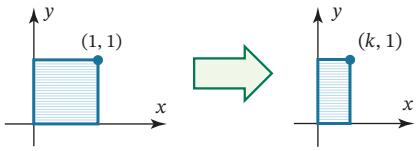
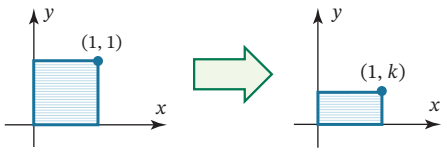
TABLE 3

Operator	Standard Matrix	Effect on the Unit Square
Rotation about the origin through a positive angle θ	$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$	

Expansions and Compressions

If the x-coordinate of each point in the plane is multiplied by a constant k where $k > 1$, the effect is to expand the unit square in the x-direction, and if the y-coordinate is multiplied by such k, then the effect is to expand the unit square in the y-direction. Operators of this type are called ***expansions***. In contrast, if $0 < k < 1$ then the effect is to compress the unit square, so such operators are called ***compressions***. Table 4 illustrates the effect on the unit square of expansions and compressions.

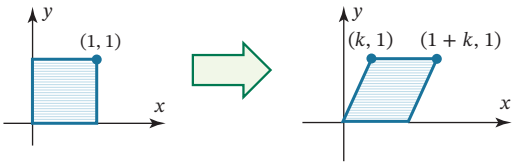
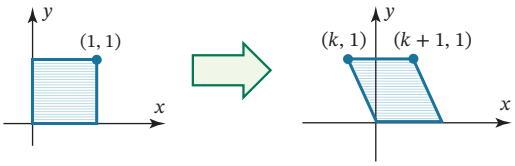
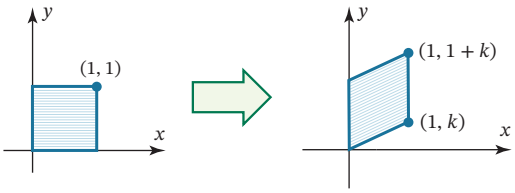
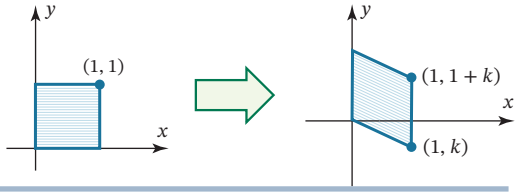
TABLE 4

Operator	Standard Matrix	Effect on the Unit Square
Expansion in the x-direction with factor k $(k > 1)$	$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$	
Expansion in the y-direction with factor k $(k > 1)$	$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$	
Compression in the x-direction with factor k $(0 < k < 1)$	$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$	
Compression in the y-direction with factor k $(0 < k < 1)$	$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$	

Shears

A matrix operator of the form $T(x, y) = (x + ky, y)$ translates a point (x, y) in the xy-plane parallel to the x-axis by an amount ky that is proportional to the y-coordinate of the point. This operator leaves the points on the x-axis fixed (since $y = 0$), but as we progress away from the x-axis, the translation distance increases. We call this operator the ***shear in the x-direction by a factor k***. Similarly, a matrix operator of the form $T(x, y) = (x, y + kx)$ is called the ***shear in the y-direction by a factor k***. Table 5, which illustrates the basic information about shears in R^2, shows that a shear is in the positive direction if $k > 0$ and the negative direction if $k < 0$.

TABLE 5

Operator	Standard Matrix	Effect on the Unit Square
Shear in the positive x-direction by a factor k $(k > 0)$	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	
Shear in the negative x-direction by a factor k $(k < 0)$	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	
Shear in the positive y-direction by a factor k $(k > 0)$	$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$	
Shear in the negative y-direction by a factor k $(k < 0)$	$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$	

EXAMPLE 3 | Transformation of the Unit Square

(*a*) Find the standard matrix for the operator on R^2 that first shears by a factor of 2 in the x-direction and then reflects the result about the line $y = x$. Sketch the image of the unit square under this operator.

(*b*) Find the standard matrix for the operator on R^2 that first reflects about $y = x$ and then shears by a factor of 2 in the x-direction. Sketch the image of the unit square under this operator.

(*c*) Confirm algebraically and visually that the shear and the reflection in parts (a) and (b) do not commute.

Solution (*a*) The standard matrix for the shear is

$$A_1 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

and for the reflection is

$$A_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Thus, the standard matrix for the shear followed by the reflection is

$$A_2A_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$$

Solution (*b*) The standard matrix for the reflection followed by the shear is

$$A_1A_2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$$

Solution (*c*) The computations in Solutions (*a*) and (*b*) show that $A_1A_2 \neq A_2A_1$, so the standard matrices, and hence the operators, do not commute. The same conclusion follows from Figures 8.6.4 and 8.6.5 since the two operators produce different images of the unit square.

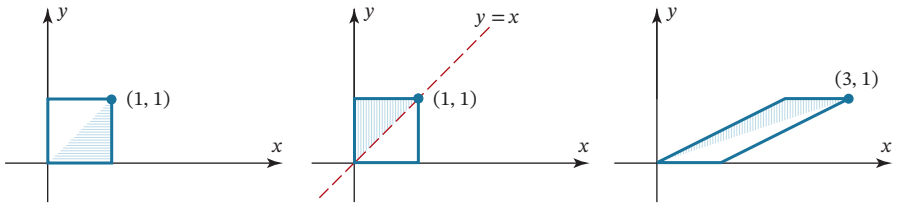


FIGURE 8.6.4

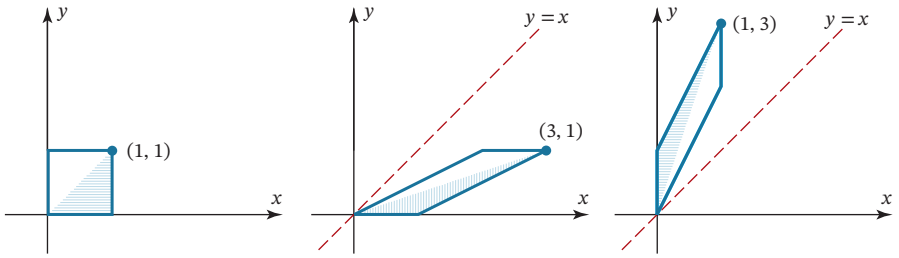


FIGURE 8.6.5

Dilations and Contractions

If k is a nonnegative scalar, then the operator $T(\mathbf{x}) = k\mathbf{x}$ on R^2 or R^3 has the effect of increasing or decreasing the length of each vector by a factor of k. If $0 \leq k < 1$ the operator is called a ***contraction*** with factor k, and if $k > 1$ it is called a ***dilation*** with factor k (Figure 8.6.6). If $k = 1$, then T is the identity operator. Table 6 illustrates the effect of contractions and dilations on the unit square in R^2— contractions shrink the square and dilations enlarge it.

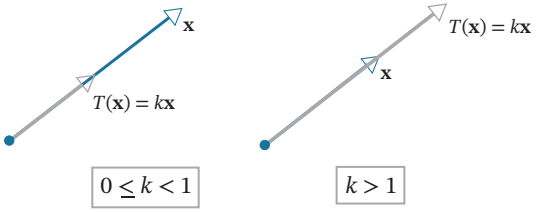
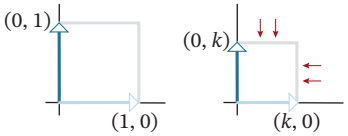
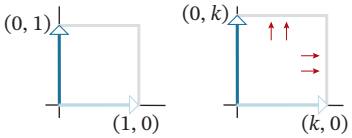


FIGURE 8.6.6

TABLE 6

Operator	Effect on the Unit Square	Standard Matrix
Contraction with factor k in R^2 $(0 \le k < 1)$		$\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$
Dilation with factor k in R^2 $(k > 1)$		

Remark Note that the standard matrix for both contractions and dilations is a diagonal matrix all of whose diagonal entries are nonnegative and the same. A matrix operator with this property is called a ***uniform scaling***.

EXAMPLE 4 | Transformations with Diagonal Matrices

Discuss the geometric effect on the unit square of multiplication by a diagonal matrix

$$A = \begin{bmatrix} k_1 & 0 \\ 0 & k_2 \end{bmatrix}$$

in which the entries k_1 and k_2 are positive real numbers ($\neq 1$).

Solution In the special case where $k_1 = k_2$, multiplication by A is either a contraction or dilation. More generally, the effect of multiplication by A can be seen by observing that this matrix can be factored as

$$A = \begin{bmatrix} k_1 & 0 \\ 0 & k_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & k_2 \end{bmatrix} \begin{bmatrix} k_1 & 0 \\ 0 & 1 \end{bmatrix}$$

which shows that multiplication by A causes a compression or expansion of the unit square by a factor of k_1 in the x-direction followed by an expansion or compression of the unit square by a factor of k_2 in the y-direction.

EXAMPLE 5 | Reflection About the Origin

As illustrated in Figure 8.6.7, multiplication by the matrix

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

has the geometric effect of reflecting the unit square about the origin. Note, however, that the matrix equation

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

together with Table 1 shows that the same result can be obtained by first reflecting the unit square about the x-axis and then reflecting that result about the y-axis. You should be able to see this as well from Figure 8.6.7.

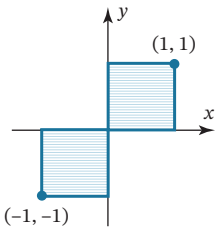


FIGURE 8.6.7

EXAMPLE 6 | Reflection About the Line $y = -x$

We leave it for you to verify that multiplication by the matrix

$$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

reflects the unit square about the line $y = -x$ (**Figure 8.6.8**).

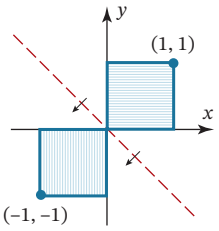


FIGURE 8.6.8

Decomposing Invertible Matrix Transformations

Our next objective is to show that all invertible matrix transformations of R^2 are expressible as compositions of compressions, expansions, reflections, and shears, so in this sense these four simple transformations are the building blocks of even the most complicated invertible matrix transformations of R^2. We begin with the following theorem.

Theorem 8.6.2

If E is an elementary matrtix, then $T_E : R^2 \rightarrow R^2$ is one of the following:

(a) A shear along a coordinate axis.
(b) A reflection about $y = x$.
(c) A compression along a coordinate axis.
(d) An expansion along a coordinate axis.
(e) A reflection about a coordinate axis.
(f) A compression or expansion along a coordinate axis followed by a reflection about a coordinate axis.

Proof Because a 2×2 elementary matrix results from performing a single elementary row operation on the 2×2 identity matrix, such a matrix must have one of the following forms (verify):

$$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$$

The first two matrices represent shears along coordinate axes, and the third represents a reflection about $y = x$. If $k > 0$, the last two matrices represent compressions or expansions along coordinate axes, depending on whether $0 \leq k < 1$ or $k > 1$. If $k < 0$, and if we express k in the form $k = -k_1$, where $k_1 > 0$, then the last two matrices can be written as

$$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -k_1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} k_1 & 0 \\ 0 & 1 \end{bmatrix} \tag{1}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -k_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & k_1 \end{bmatrix} \tag{2}$$

Since $k_1 > 0$, the product in (1) represents a compression or expansion along the x-axis followed by a reflection about the y-axis, and (2) represents a compression or expansion along the y-axis followed by a reflection about the x-axis. In the case where $k = -1$, transformations (1) and (2) are simply reflections about the y-axis and x-axis, respectively. ■

We know from Theorem 1.5.3(d) that an invertible matrix can be expressed as a product of elementary matrices, so Theorem 8.6.2 implies the following result.

Theorem 8.6.3

If $T_A : R^2 \rightarrow R^2$ is multiplication by an invertible matrix A, then the geometric effect of T_A is the same as an appropriate succession of shears, compressions, expansions, and reflections.

The next example will illustrate how Theorems 8.6.2 and 8.6.3 together with Tables 1 through 5 can be used to analyze the geometric effect of multiplication by a 2×2 invertible matrix.

EXAMPLE 7 | Decomposing a Matrix Operator

In Example 2 we illustrated the effect on the unit square of multiplication by

$$A = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$$

(see **Figure 8.6.3**). Express this matrix as a product of elementary matrices, and then describe the effect of multiplication by the matrix A in terms of shears, compressions, expansions, and reflections.

Solution The matrix A can be reduced to the identity matrix as follows:

$$\begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Interchange the first and second rows. Multiply the first row by $\frac{1}{2}$. Add $-\frac{1}{2}$ times the second row to the first.

These three successive row operations can be performed by multiplying A on the left successively by

$$E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad E_2 = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & 1 \end{bmatrix}$$

Inverting these matrices and using Formula (4) of Section 1.5 yields

$$A = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} = E_1^{-1}E_2^{-1}E_3^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}$$

Reading from right to left we can now see that the geometric effect of multiplying by A is equivalent to successively

1. shearing by a factor of $\frac{1}{2}$ in the x-direction;
2. expanding by a factor of 2 in the x-direction;
3. reflecting about the line $y = x$.

This is illustrated in **Figure 8.6.9**, whose end result agrees with that in Example 2.

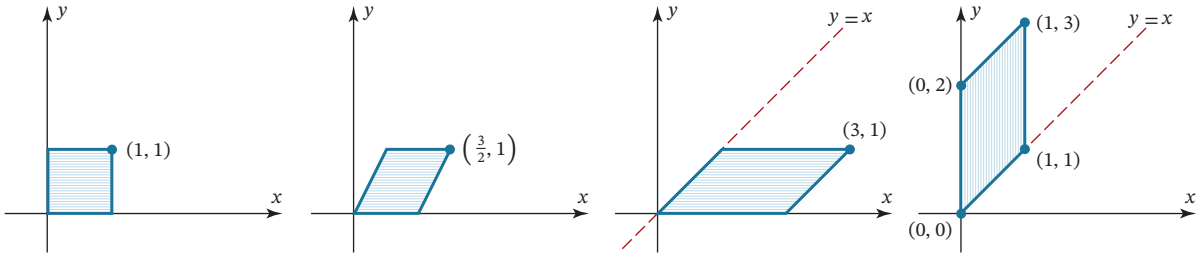
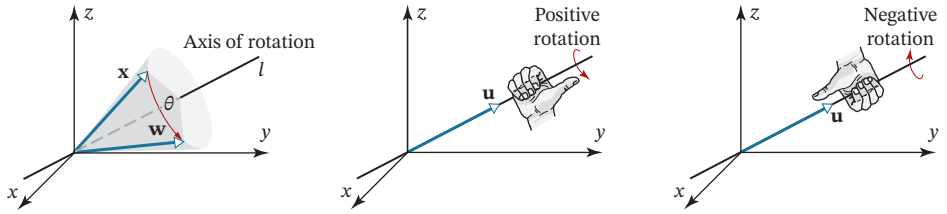


FIGURE 8.6.9

Remark Translations in R^2 are important operations in computer graphics, but they are not linear transformations (Example 8 of Section 8.1) and hence cannot be accomplished using 2×2 matrices as multipliers. A method for performing translations using 3×3 matrices is discussed in Supplementary Exercise 26.

Rotations in R^3

A rotation of vectors in R^3 is commonly described in relation to a line through the origin called the ***axis of rotation*** and a unit vector $\mathbf{u}$ along that line (**Figure 8.6.10*a***). The unit vector and what is called the ***right-hand rule*** can be used to establish a sign for the angle of rotation by cupping the fingers of your right hand so they curl in the direction of rotation and observing the direction of your thumb. If your thumb points in the direction of $\mathbf{u}$, then the angle of rotation is regarded to be ***positive*** relative to $\mathbf{u}$, and if it points in the direction opposite to $\mathbf{u}$, then it is regarded to be ***negative*** relative to $\mathbf{u}$ (**Figure 8.6.10*b***).



(*a*) Angle of rotation (*b*) Right-hand rule

FIGURE 8.6.10

For rotations about the coordinate axes in R^3, we will take the unit vectors to be $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, in which case an angle of rotation will be positive if it is counterclockwise looking toward the origin along the positive coordinate axis and will be negative if it is clockwise. **Table 7** shows the standard matrices for the ***rotation operators*** on R^3 that rotate each vector about one of the coordinate axes through an angle θ. You will find it instructive to compare these matrices to those in Table 5 of Section 1.8.

Yaw, Pitch, and Roll

In aeronautics and astronautics, the orientation of an aircraft or space shuttle relative to an xyz-coordinate system is often described in terms of angles called ***yaw***, ***pitch***, and ***roll***. If, for example, an aircraft is flying along the y-axis and the xy-plane defines the horizontal, then the aircraft's angle of rotation about the z-axis is called the ***yaw***, its angle of rotation about the x-axis is called the ***pitch***, and its angle of rotation about the y-axis is called the ***roll***. A combination of yaw, pitch, and roll can be achieved by a single rotation about some axis through the origin. This is, in fact, how a space shuttle makes attitude adjustments—it doesn't perform each rotation separately; it calculates one axis, and rotates about that axis to get the correct orientation. Such rotation maneuvers are used to align an antenna, point the nose toward a celestial object, or position a payload bay for docking.

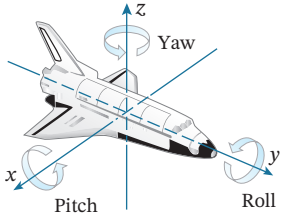
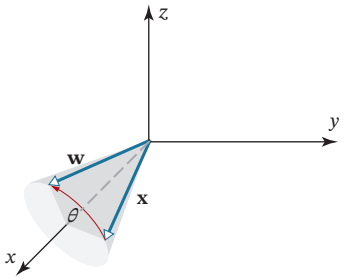
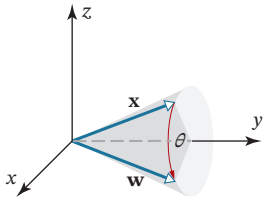
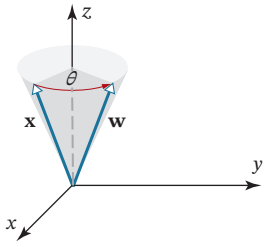


TABLE 7

Operator	Illustration	Rotation Equations	Standard Matrix
Counterclockwise rotation about the positive x-axis through an angle θ		$w_1 = x$ $w_2 = y\cos\theta - z\sin\theta$ $w_3 = y\sin\theta + z\cos\theta$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$
Counterclockwise rotation about the positive y-axis through an angle θ		$w_1 = x\cos\theta + z\sin\theta$ $w_2 = y$ $w_3 = -x\sin\theta + z\cos\theta$	$\begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$
Counterclockwise rotation about the positive z-axis through an angle θ		$w_1 = x\cos\theta - y\sin\theta$ $w_2 = x\sin\theta + y\cos\theta$ $w_3 = z$	$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

For completeness, we note that the standard matrix for a counterclockwise rotation through an angle θ about an axis in R^3, which is determined by an arbitrary *unit vector* $\mathbf{u} = (a, b, c)$ that has its initial point at the origin, is

$$\begin{bmatrix} a^2(1-\cos\theta)+\cos\theta & ab(1-\cos\theta)-c\sin\theta & ac(1-\cos\theta)+b\sin\theta \\ ab(1-\cos\theta)+c\sin\theta & b^2(1-\cos\theta)+\cos\theta & bc(1-\cos\theta)-a\sin\theta \\ ac(1-\cos\theta)-b\sin\theta & bc(1-\cos\theta)+a\sin\theta & c^2(1-\cos\theta)+\cos\theta \end{bmatrix} \tag{3}$$

The derivation can be found in the book *Principles of Interactive Computer Graphics*, by W. M. Newman and R. F. Sproull (New York: McGraw-Hill, 1979). You may find it instructive to derive the results in Table 7 as special cases of this more general result.

Exercise Set 8.6

1. Use the method of Example 1 to find an equation for the image of the line $y = 4x$ under multiplication by the matrix

$$A = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$$

2. Use the method of Example 1 to find an equation for the image of the line $y = -4x + 3$ under multiplication by the matrix

$$A = \begin{bmatrix} 4 & -3 \\ 3 & -2 \end{bmatrix}$$

In Exercises 3–4, find an equation for the image of the line $y = 2x$ that results from the stated transformation.

3. A shear by a factor 3 in the x-direction.

4. A compression with factor $\frac{1}{2}$ in the y-direction.

In Exercises 5–6, sketch the image of the unit square under multiplication by the given invertible matrix. As in Example 2, number the edges of the unit square and its image so it is clear how those edges correspond.

5. $\begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix}$

6. $\begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$

In each part of Exercises 7–8, find the standard matrix for a single operator that performs the stated succession of operations.

7. **a.** Compresses by a factor of $\frac{1}{2}$ in the x-direction, then expands by a factor of 5 in the y-direction.

b. Expands by a factor of 5 in the y-direction, then shears by a factor of 2 in the y-direction.

c. Reflects about $y = x$, then rotates through an angle of 180° about the origin.

8. **a.** Reflects about the y-axis, then expands by a factor of 5 in the x-direction, and then reflects about $y = x$.

b. Rotates through 30° about the origin, then shears by a factor of -2 in the y-direction, and then expands by a factor of 3 in the y-direction.

In each part of Exercises 9–10, determine whether the stated operators commute.

9. **a.** A reflection about the x-axis and a compression in the x-direction with factor $\frac{1}{3}$.

b. A reflection about the line $y = x$ and an expansion in the x-direction with factor 2.

10. **a.** A shear in the y-direction by a factor $\frac{1}{4}$ and a shear in the y-direction by a factor $\frac{3}{5}$.

b. A shear in the y-direction by a factor $\frac{1}{4}$ and a shear in the x-direction by a factor $\frac{3}{5}$.

In Exercises 11–14, express the matrix as a product of elementary matrices, and then describe the effect of multiplication by A in terms of shears, compressions, expansions, and reflections.

11. $A = \begin{bmatrix} 4 & 4 \\ 0 & -2 \end{bmatrix}$

12. $A = \begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix}$

13. $A = \begin{bmatrix} 0 & -2 \\ 4 & 0 \end{bmatrix}$

14. $A = \begin{bmatrix} 1 & -3 \\ 4 & 6 \end{bmatrix}$

In each part of Exercises 15–16, describe, in words, the effect on the unit square of multiplication by the given diagonal matrix.

15. **a.** $A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$ **b.** $A = \begin{bmatrix} 1 & 0 \\ 0 & -5 \end{bmatrix}$

16. **a.** $A = \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix}$ **b.** $A = \begin{bmatrix} -3 & 0 \\ 0 & -1 \end{bmatrix}$

17. **a.** Show that multiplication by

$$A = \begin{bmatrix} 3 & 1 \\ 6 & 2 \end{bmatrix}$$

maps each point in the plane onto the line $y = 2x$.

b. It follows from part (a) that the noncollinear points $(1, 0)$, $(0, 1)$, $(-1, 0)$ are mapped onto a line. Does this violate part (e) of Theorem 8.6.1?

18. Find the matrix for a shear in the x-direction that transforms the triangle with vertices $(0, 0)$, $(2, 1)$, and $(3, 0)$ into a right triangle with the right angle at the origin.

19. In accordance with part (c) of Theorem 8.6.1, show that multiplication by the invertible matrix

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

maps the parallel lines $y = 3x + 1$ and $y = 3x - 2$ into parallel lines.

20. Draw a figure that shows the image of the triangle with vertices $(0, 0)$, $(1, 0)$, and $(0.5, 1)$ under a shear by a factor of 2 in the x-direction.

21. **a.** Draw a figure that shows the image of the triangle with vertices $(0, 0)$, $(1, 0)$, and $(0.5, 1)$ under multiplication by

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

b. Find a succession of shears, compressions, expansions, and reflections that produces the same image.

22. Find the endpoints of the line segment that results when the line segment from $P(1, 2)$ to $Q(3, 4)$ is transformed by

a. a compression with factor $\frac{1}{2}$ in the y-direction.

b. a rotation of 30° about the origin.

23. Draw a figure showing the italicized letter "T" that results when the letter in the accompanying figure is sheared by a factor $\frac{1}{4}$ in the x-direction.

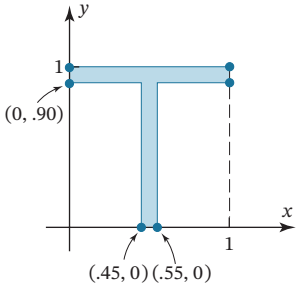


FIGURE Ex-23

24. Can an invertible matrix operator on R^2 map a square region into a triangular region? Justify your answer.

25. Find the image of the triangle with vertices $(0, 0)$, $(1, 1)$, $(2, 0)$ under multiplication by

$$A = \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix}$$

Does your answer violate part (e) of Theorem 8.6.1? Explain.

26. In R^3 the ***shear in the xy-direction by a factor k*** is the matrix transformation that moves each point (x, y, z) parallel to the xy-plane to the new position $(x + kz, y + kz, z)$. (See the accompanying figure.)

a. Find the standard matrix for the shear in the xy-direction by a factor k.

b. How would you define the shear in the xz-direction by a factor k and the shear in the yz-direction by a factor k? What are the standard matrices for these matrix transformations?

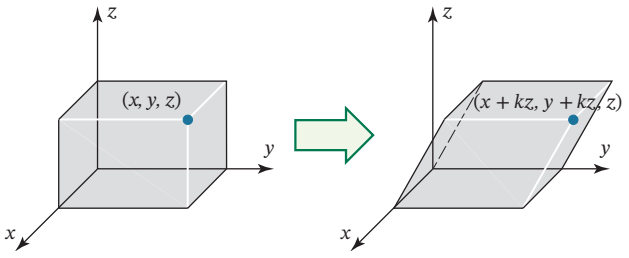


FIGURE Ex-26

In Exercises 27–28, find the standard matrix for the operator $T : R^3 \rightarrow R^3$ that performs the stated rotation.

27. **a.** rotates each vector 30° counterclockwise about the z-axis (looking along the positive z-axis toward the origin).

b. rotates each vector 45° clockwise about the x-axis (looking along the positive x-axis toward the origin).

28. **a.** rotates each vector 90° counterclockwise about the y-axis (looking along the positive y-axis toward the origin).

b. rotates each vector 90° clockwise about the positive z-axis looking toward the origin.

29. Use Formula (3) to find the standard matrix for a rotation of 180° about the axis determined by the vector $\mathbf{v} = (2, 2, 1)$. [*Note:* Formula (3) requires that the vector defining the axis of rotation have length 1.]

30. Use Formula (3) to find the standard matrix for a rotation of $\pi/2$ radians about the axis determined by $\mathbf{v} = (1, 1, 1)$. [*Note:* Formula (3) requires that the vector defining the axis of rotation have length 1.]

31. Use Formula (3) to derive the standard matrices for the rotations about the x-axis, the y-axis, and the z-axis through an angle of 90° in R^3.

32. ***Euler's Axis of Rotation Theorem*** states: *If A is an orthogonal 3×3 matrix for which $\det(A) = 1$, then multiplication by A is a rotation about a line through the origin in R^3. Moreover, if $\mathbf{u}$ is a unit vector along this line, then $A\mathbf{u} = \mathbf{u}$.*

a. Confirm that the following matrix A is orthogonal, that $\det(A) = 1$, and that there is a unit vector $\mathbf{u}$ for which $A\mathbf{u} = \mathbf{u}$.

$$A = \begin{bmatrix} \frac{2}{7} & \frac{3}{7} & \frac{6}{7} \\ \frac{3}{7} & -\frac{6}{7} & \frac{2}{7} \\ \frac{6}{7} & \frac{2}{7} & -\frac{3}{7} \end{bmatrix}$$

b. Use Formula (3) to prove that if A is a 3×3 orthogonal matrix for which $\det(A) = 1$, then the angle of rotation resulting from multiplication by A satisfies the equation $\cos\theta = \frac{1}{2}[\text{tr}(A) - 1]$. Use this result to find the angle of rotation for the rotation matrix in part (a).

Working with Proofs

33. Prove part (a) of Theorem 8.6.1. [*Hint:* A line in the plane has an equation of the form $Ax + By + C = 0$, where A and B are not both zero. Use the method of Example 1 to show that the image of this line under multiplication by the invertible matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

has the equation $A'x + B'y + C = 0$, where

$$A' = (dA - cB)/(ad - bc)$$

and

$$B' = (-bA + aB)/(ad - bc)$$

Then show that A' and B' are not both zero to conclude that the image is a line.

34. Use the hint in Exercise 33 to prove parts (b) and (c) of Theorem 8.6.1.

True-False Exercises

TF. In parts **(a)–(g)** determine whether the statement is true or false, and justify your answer.

a. The image of the unit square under a one-to-one matrix operator is a square.

b. A 2×2 invertible matrix operator has the geometric effect of a succession of shears, compressions, expansions, and reflections.

c. The image of a line under an invertible matrix operator is a line.

d. Every reflection operator on R^2 is its own inverse.

e. The matrix $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ represents reflection about a line.

f. The matrix $\begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$ represents a shear.

g. The matrix $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ represents an expansion.

Working with Technology

T1. **a.** Find the standard matrix for the linear operator on R^3 that performs a counterclockwise rotation of 47° about the x-axis, followed by a counterclockwise rotation of 68° about the y-axis, followed by a counterclockwise rotation of 33° about the z-axis.

b. Find the image of the point $(1, 1, 1)$ under the operator in part (a).

T2. Find the standard matrix for the linear operator on R^2 that first reflects each point in the plane about the line through the origin that makes an angle of 27° with the positive x-axis and then projects the resulting point orthogonally onto the line through the origin that makes an angle of 51° with the positive x-axis.

Chapter 8 Supplementary Exercises

1. Let A be an $n \times n$ matrix, B a nonzero $n \times 1$ matrix, and $\mathbf{x}$ a vector in R^n expressed in matrix notation. Is $T(\mathbf{x}) = A\mathbf{x} + B$ a linear operator on R^n? Justify your answer.

2. Let

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

a. Show that

$$A^2 = \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix} \quad \text{and} \quad A^3 = \begin{bmatrix} \cos 3\theta & -\sin 3\theta \\ \sin 3\theta & \cos 3\theta \end{bmatrix}$$

b. Based on your answer to part (a), make a guess at the form of the matrix A^n for any positive integer n.

c. By considering the geometric effect of multiplication by A, obtain the result in part (b) geometrically.

3. Devise a method for finding two $n \times n$ matrices that are not similar. Use your method to find two 3×3 matrices that are not similar.

4. Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ be fixed vectors in R^n, and let $T : R^n \to R^m$ be the function defined by $T(\mathbf{x}) = (\mathbf{x} \cdot \mathbf{v}_1, \mathbf{x} \cdot \mathbf{v}_2, \ldots, \mathbf{x} \cdot \mathbf{v}_m)$, where $\mathbf{x} \cdot \mathbf{v}_i$ is the Euclidean inner product on R^n.

a. Show that T is a linear transformation.

b. Show that the matrix with row vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ is the standard matrix for T.

5. Let $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4\}$ be the standard basis for the vector space R^4, and let $T : R^4 \to R^3$ be the linear transformation for which

$$T(\mathbf{e}_1) = (1, 2, 1), \quad T(\mathbf{e}_2) = (0, 1, 0),$$
$$T(\mathbf{e}_3) = (1, 3, 0), \quad T(\mathbf{e}_4) = (1, 1, 1)$$

a. Find bases for the range and kernel of T.

b. Find the rank and nullity of T.

6. Suppose that vectors in R^3 are denoted by 1×3 matrices, and define $T : R^3 \to R^3$ by

$$T([x_1 \quad x_2 \quad x_3]) = [x_1 \quad x_2 \quad x_3] \begin{bmatrix} -1 & 2 & 4 \\ 3 & 0 & 1 \\ 2 & 2 & 5 \end{bmatrix}$$

a. Find a basis for the kernel of T.

b. Find a basis for the range of T.

7. Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ be a basis for a vector space V, and let $T : V \to V$ be the linear operator for which

$$T(\mathbf{v}_1) = \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 + 3\mathbf{v}_4$$
$$T(\mathbf{v}_2) = \mathbf{v}_1 - \mathbf{v}_2 + 2\mathbf{v}_3 + 2\mathbf{v}_4$$
$$T(\mathbf{v}_3) = 2\mathbf{v}_1 - 4\mathbf{v}_2 + 5\mathbf{v}_3 + 3\mathbf{v}_4$$
$$T(\mathbf{v}_4) = -2\mathbf{v}_1 + 6\mathbf{v}_2 - 6\mathbf{v}_3 - 2\mathbf{v}_4$$

a. Find the rank and nullity of T.

b. Determine whether T is one-to-one.

8. Let V and W be vector spaces, let T, T_1, and T_2 be linear transformations from V to W, and let k be a scalar. Define new transformations, $T_1 + T_2$ and kT, by the formulas

$$(T_1 + T_2)(\mathbf{x}) = T_1(\mathbf{x}) + T_2(\mathbf{x})$$
$$(kT)(\mathbf{x}) = k(T(\mathbf{x}))$$

a. Show that $(T_1 + T_2) : V \to W$ and $kT : V \to W$ are both linear transformations.

b. Show that the set of all linear transformations from V to W with the operations in part (a) is a vector space.

9. Let A and B be similar matrices. Prove:

a. A^T and B^T are similar.

b. If A and B are invertible, then A^{-1} and B^{-1} are similar.

10. ***(Fredholm Alternative Theorem)*** Let $T : V \to V$ be a linear operator on an n-dimensional vector space. Prove that *exactly one* of the following statements holds:

i. The equation $T(\mathbf{x}) = \mathbf{b}$ has a solution for all vectors $\mathbf{b}$ in V.

ii. Nullity of $T > 0$.

11. Let $T : M_{22} \to M_{22}$ be the linear operator defined by

$$T(X) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} X + X \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

Find the rank and nullity of T.

12. Prove: If A and B are similar matrices, and if B and C are also similar matrices, then A and C are similar matrices.

13. Let $L : M_{22} \to M_{22}$ be the linear operator that is defined by $L(M) = M^T$. Find the matrix for L with respect to the standard basis for M_{22}.

14. Let $B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ and $B' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be bases for a vector space V, and let

$$P = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 1 & 4 \\ 0 & 1 & 2 \end{bmatrix}$$

be the transition matrix from B' to B.

a. Express $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ as linear combinations of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$.

b. Express $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ as linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$.

15. Let $B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be a basis for a vector space V, and let $T : V \to V$ be a linear operator for which

$$[T]_B = \begin{bmatrix} -3 & 4 & 7 \\ 1 & 0 & -2 \\ 0 & 1 & 0 \end{bmatrix}$$

Find $[T]_{B'}$, where $B' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is the basis for V defined by

$$\mathbf{v}_1 = \mathbf{u}_1, \quad \mathbf{v}_2 = \mathbf{u}_1 + \mathbf{u}_2, \quad \mathbf{v}_3 = \mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3$$

16. Show that the matrices

$$\begin{bmatrix} 1 & 1 \\ -1 & 4 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

are similar but that

$$\begin{bmatrix} 3 & 1 \\ -6 & -2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}$$

are not.

17. Suppose that $T : V \to V$ is a linear operator, and B is a basis for V for which

$$[T(\mathbf{x})]_B = \begin{bmatrix} x_1 - x_2 + x_3 \\ x_2 \\ x_1 - x_3 \end{bmatrix} \quad \text{if} \quad [\mathbf{x}]_B = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Find $[T]_B$.

18. Let $T : V \to V$ be a linear operator. Prove that T is one-to-one if and only if $\det(T) \neq 0$.

19. ***(Calculus required)***

a. Show that if $\mathbf{f} = f(x)$ is twice differentiable, then the function $D : C^2(-\infty, \infty) \to F(-\infty, \infty)$ defined by the formula $D(\mathbf{f}) = f''(x)$ is a linear transformation.

b. Find a basis for the kernel of D.

c. Show that the set of functions satisfying the equation $D(\mathbf{f}) = f(x)$ is a two-dimensional subspace of $C^2(-\infty, \infty)$, and find a basis for this subspace.

20. Let $T : P_2 \to R^3$ be the function defined by the formula

$$T(p(x)) = \begin{bmatrix} p(-1) \\ p(0) \\ p(1) \end{bmatrix}$$

a. Find $T(x^2 + 5x + 6)$.

b. Show that T is a linear transformation.

c. Show that T is one-to-one.

d. Find $T^{-1}(0, 3, 0)$.

e. Sketch the graph of the polynomial in part (d).

21. Let x_1, x_2, and x_3 be distinct real numbers such that

$$x_1 < x_2 < x_3$$

and let $T : P_2 \to R^3$ be the function defined by the formula

$$T(p(x)) = \begin{bmatrix} p(x_1) \\ p(x_2) \\ p(x_3) \end{bmatrix}$$

a. Show that T is a linear transformation.

b. Show that T is one-to-one.

c. Verify that if a_1, a_2, and a_3 are any real numbers, then

$$T^{-1}\left(\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}\right) = a_1P_1(x) + a_2P_2(x) + a_3P_3(x)$$

where

$$P_1(x) = \frac{(x - x_2)(x - x_3)}{(x_1 - x_2)(x_1 - x_3)}$$

$$P_2(x) = \frac{(x - x_1)(x - x_3)}{(x_2 - x_1)(x_2 - x_3)}$$

$$P_3(x) = \frac{(x - x_1)(x - x_2)}{(x_3 - x_1)(x_3 - x_2)}$$

d. What relationship exists between the graph of the function

$$a_1P_1(x) + a_2P_2(x) + a_3P_3(x)$$

and the points (x_1, a_1), (x_2, a_2), and (x_3, a_3)?

22. ***(Calculus required)*** Let $p(x)$ and $q(x)$ be continuous functions, and let V be the subspace of $C(-\infty, \infty)$ consisting of all twice differentiable functions. Define $L : V \to V$ by

$$L(y(x)) = y''(x) + p(x)y'(x) + q(x)y(x)$$

a. Show that L is a linear transformation.

b. Consider the special case where $p(x) = 0$ and $q(x) = 1$. Show that the function

$$\phi(x) = c_1 \sin x + c_2 \cos x$$

is in the kernel of L for all real values of c_1 and c_2.

23. ***(Calculus required)*** Let $D : P_n \to P_n$ be the differentiation operator $D(\mathbf{p}) = \mathbf{p}'$. Show that the matrix for D relative to the basis $B = \{1, x, x^2, \ldots, x^n\}$ is

$$\begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 2 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & 0 & \cdots & n \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

24. ***(Calculus required)*** It can be shown that for any real number c, the vectors

$$1, \quad x - c, \quad \frac{(x - c)^2}{2!}, \ldots, \quad \frac{(x - c)^n}{n!}$$

form a basis for P_n. Find the matrix for the differentiation operator of Exercise 23 with respect to this basis.

25. ***(Calculus required)*** Let $J : P_n \to P_{n+1}$ be the integration transformation defined by

$$J(\mathbf{p}) = \int_0^x (a_0 + a_1t + \cdots + a_nt^n)\,dt$$

$$= a_0x + \frac{a_1}{2}x^2 + \cdots + \frac{a_n}{n+1}x^{n+1}$$

where $\mathbf{p} = a_0 + a_1x + \cdots + a_nx^n$. Find the matrix for J with respect to the standard bases for P_n and P_{n+1}.

26. This exercise illustrates a method for using a matrix transformation in R^3 to translate a point (x, y) in R^2 to a point $(x + x_0,\ y + y_0)$.

a. Let

$$\mathbf{v} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \text{ and } \mathbf{w} = \begin{bmatrix} x + x_0 \\ y + y_0 \\ 1 \end{bmatrix}$$

Find a 3×3 matrix M for which $M\mathbf{v} = \mathbf{w}$. The first top two entries in $\mathbf{w}$ are the coordinates of the translated point.

b. Use the result in part (a) to find a 3×3 matrix M that translates the point (2, 1) to the point (3, 4).